

Polarization and its measurement

Samuel Martin-Gutierrez

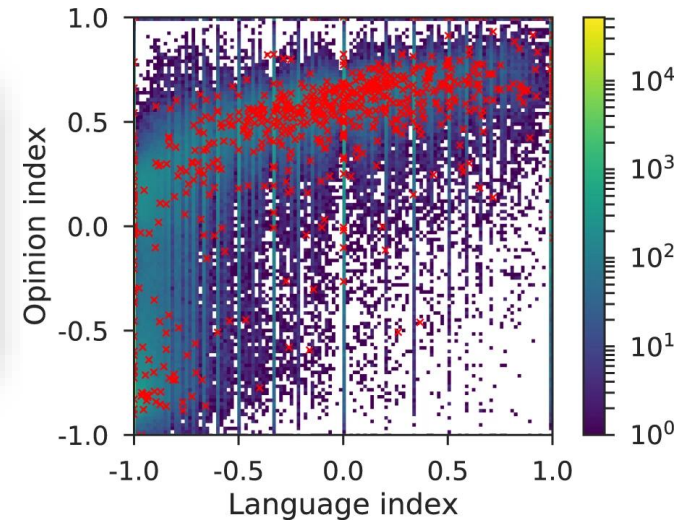
Complexity Science Hub Vienna

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Relationship between ideology and language in the Catalan independence context

[Julia Atienza-Barthelemy](#), [Samuel Martin-Gutierrez](#), [Juan C. Losada](#) & [Rosa M. Benito](#) 

[Scientific Reports](#) **9**, Article number: 17148 (2019) | [Cite this article](#)



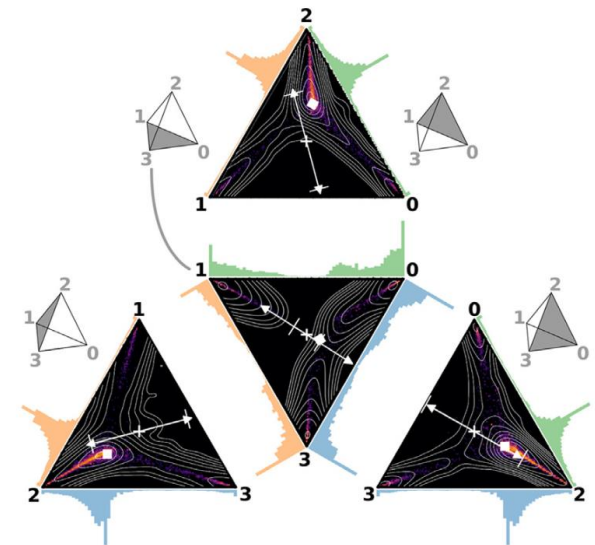
Chaos, Solitons & Fractals

Volume 169, April 2023, 113244



Multipolar social systems: Measuring polarization beyond dichotomous contexts

[Samuel Martin-Gutierrez](#)^{a,b}, [Juan C. Losada](#)^a, [Rosa M. Benito](#)^a  



What is polarization?

WHAT IS POLARIZATION?

Given polarization's prominence in contemporary political discourse, the literature provides strikingly little guidance in defining it.⁴

DiMaggio et al. *American Journal of Sociology* **102**, 690–755 (1996).

What is polarization?

- How can we define polarization?
- How do we measure polarization?
- What kind of data do we need?
- Write down an algorithm or equation to measure it.
- How do we know that it measures what we intend to measure?

What is polarization?

- Attitudinal polarization. Several aspects:
 - Extremism.
 - Fragmentation.
 - Correlation.
 - ...
- Affective polarization (in-group / out-group sentiment).
- Network / relational polarization (echo chambers).

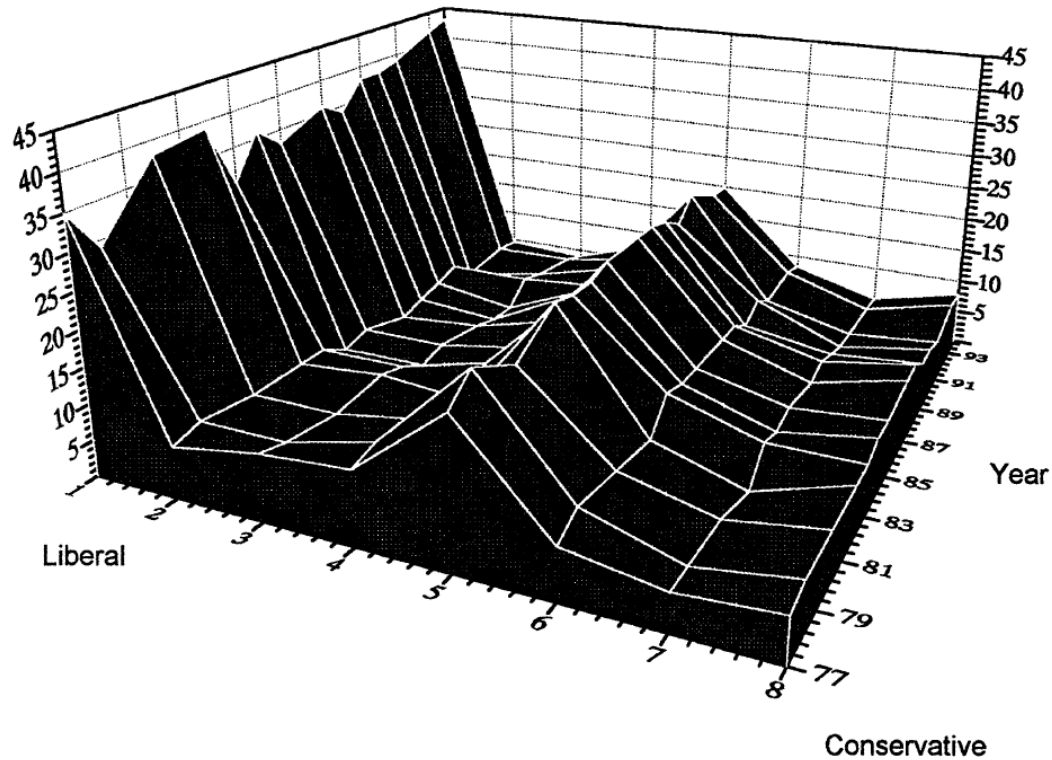


FIG. 4.—Distributions by year, attitudes toward abortion, full sample, GSS 1977–94.

Attitudinal polarization

DiMaggio et al. *American Journal of Sociology* **102**, 690–755 (1996).

What is polarization?

WHAT IS POLARIZATION?

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One-dimensional
aspects

Two of these intuitions refer to properties of single distributions:

1. Other things being equal, the more dispersed opinion becomes, the more difficult it will be for the political system to establish and maintain centrist political consensus (the **dispersion principle**).

2. Other things being equal, the greater the extent to which opinions move toward separate modes (and the more separate those modes become), the more likely it is that social conflict will ensue (the **bimodality principle**; see Esteban and Ray 1994).

Two other intuitions refer to relationships *among* distributions:

3. Other things being equal, the more closely associated different social attitudes become (both within and across opinion domains), the greater the likelihood of implacable conflict (the **constraint principle**; see Converse 1964).

4. Other things being equal, the greater the extent to which social attitudes become correlated with salient individual characteristics or identities, the more likely it is that they will become the foci of social conflict (the **consolidation principle**; see Blau 1977).

Multidimensional
aspects (correlation)

DiMaggio et al. *American Journal of Sociology* **102**, 690–755 (1996).

Measures of polarization for 1D opinion distributions

THE JOURNAL OF MATHEMATICAL SOCIOLOGY
2016, VOL. 40, NO. 2, 80–111
<http://dx.doi.org/10.1080/0022250X.2016.1147443>

 **Routledge**
Taylor & Francis Group

Disambiguation of social polarization concepts and measures

Aaron Bramson^{a,b}, Patrick Grim^c, Daniel J. Singer^d, Steven Fisher^{e,f}, William Berger^{e,f},
Graham Sack^g, and Carissa Flocken^h

ABSTRACT

This article distinguishes nine senses of polarization and provides formal measures for each one to refine the methodology used to describe polarization in distributions of attitudes. Each distinct concept is explained through a definition, formal measures, examples, and references.

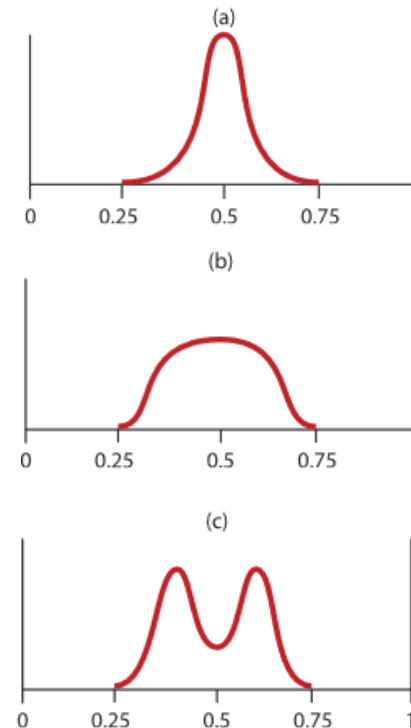
Measures of polarization for 1D opinion distributions

1. Spread
2. Dispersion
3. Coverage
4. Regionalization
5. Community fragmentation
6. Distinctness
7. Group divergence
8. Group consensus
9. Size parity

Bramson, A. *et al.* *The Journal of Mathematical Sociology* **40**, 80–111 (2016).

Measures of polarization for 1D opinion distributions

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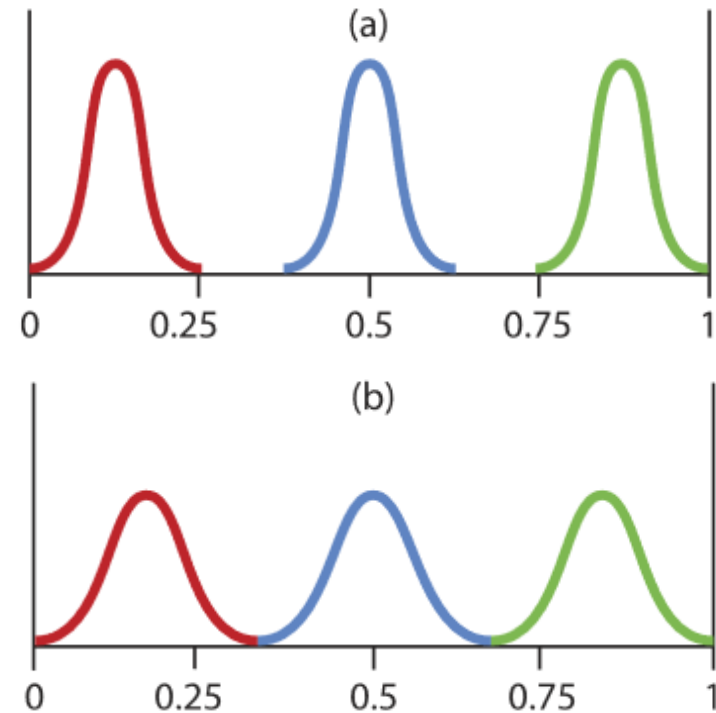


$$\text{dispersion} = \frac{2}{N} \sum_{x_i}^N |x_i - \bar{X}|,$$

Bramson, A. et al. *The Journal of Mathematical Sociology* **40**, 80–111 (2016).

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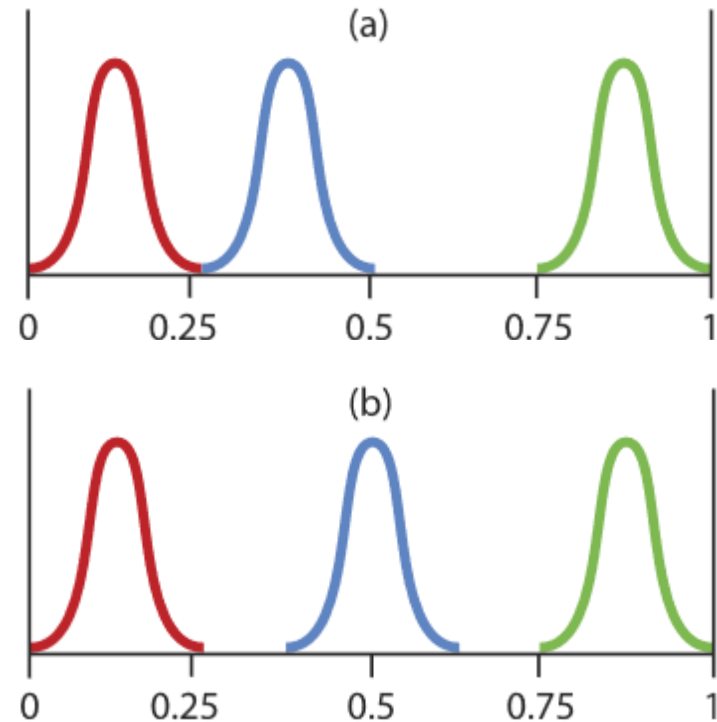


Bramson, A. et al. *The Journal of Mathematical Sociology* **40**, 80–111 (2016).

$$\text{coverage}_d = \frac{1}{|R|} \sum_{r \in R} \begin{cases} 1 & y(x_r) > 0 \\ 0 & y(x_r) = 0 \end{cases}$$

Measures of polarization for 1D opinion distributions

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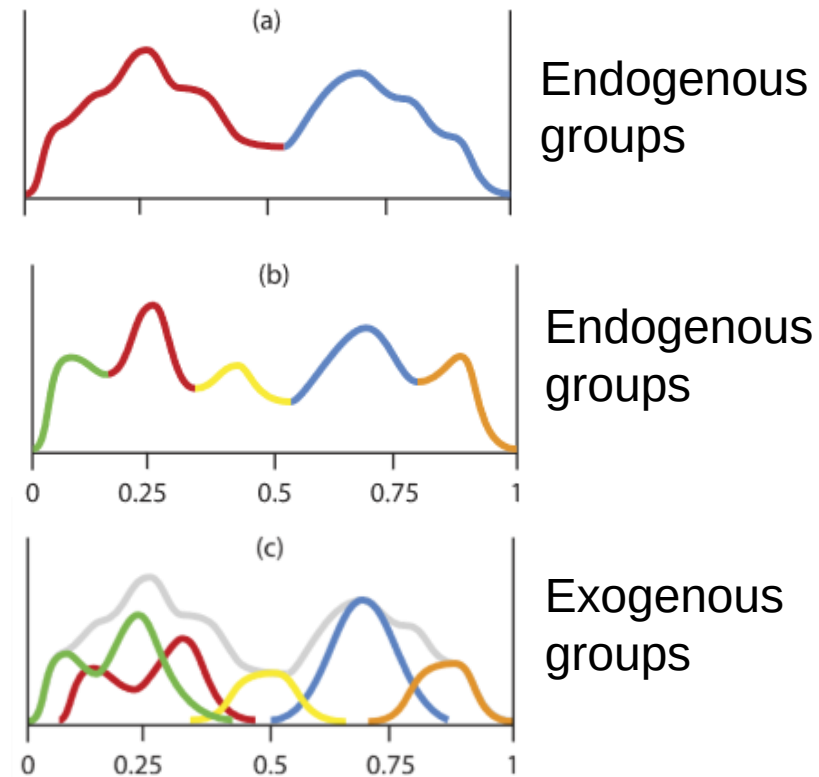


$$\sum_{r_i \in R} \begin{cases} 1 & y(r_0) = 0 \\ 1 & y(r_i) > 0 \text{ and } y(r_{i+1}) = 0 \\ 0 & \text{otherwise} \end{cases}$$

Bramson, A. et al. *The Journal of Mathematical Sociology* **40**, 80–111 (2016).

Measures of polarization for 1D opinion distributions

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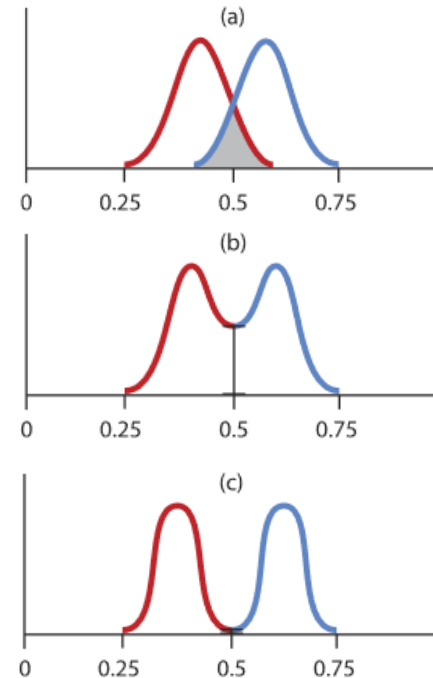


Bramson, A. et al. *The Journal of Mathematical Sociology* **40**, 80–111 (2016).

The more groups there are, the greater the polarization.

Measures of polarization for 1D opinion distributions

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$$\frac{1}{|A| + |B|} \left(1 - \sum_{r \in R} \min\{y_A(r), y_B(r)\} \right).$$

Exogenous

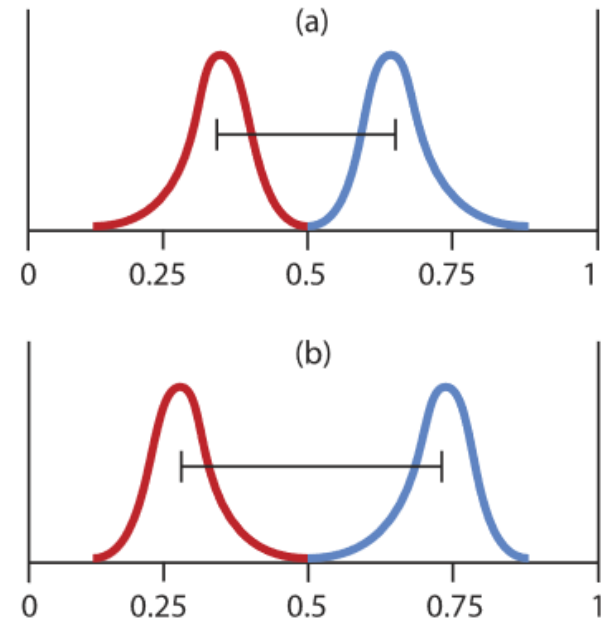
Bramson, A. et al. *The Journal of Mathematical Sociology* **40**, 80–111 (2016).

$$1 - \sum_{r \in R} \begin{cases} y(r) & y(r) < y(r-1) \text{ \& } y(r) < y(r+1) \\ 0 & \text{otherwise} \end{cases}$$

Endogenous

Measures of polarization for 1D opinion distributions

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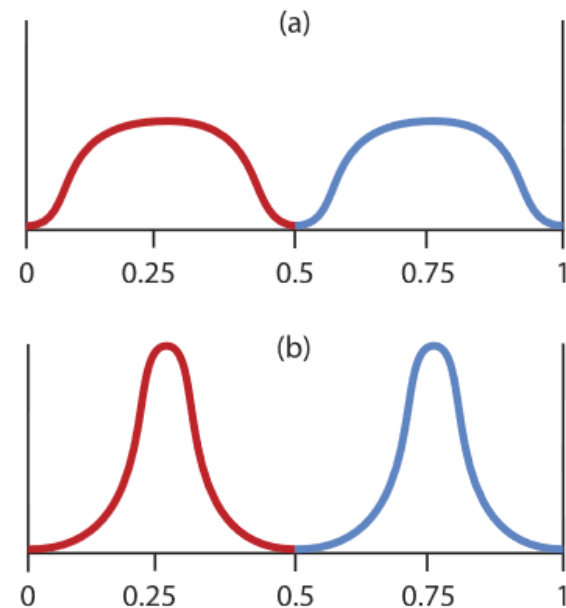


$$\text{divergence} = \frac{2}{G^2 - G} \sum_{i < j}^G |\bar{g}_i - \bar{g}_j|.$$

Bramson, A. et al. *The Journal of Mathematical Sociology* **40**, 80–111 (2016).

Measures of polarization for 1D opinion distributions

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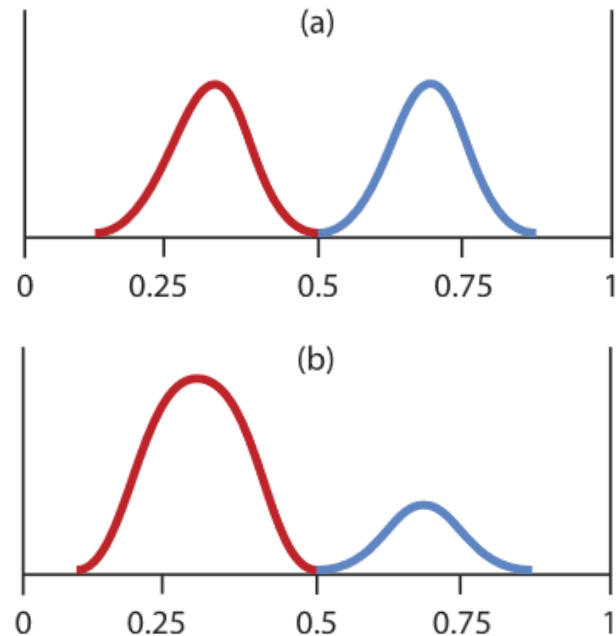
$$\text{group consensus} = 1 - \frac{1}{G} \sum_{i=1}^G \frac{2}{n_i} \sum_{j=1}^{n_i} |x_j - \bar{X}_i|.$$

Bramson, A. et al. *The Journal of Mathematical Sociology* **40**, 80–111 (2016).

Average of within-group dispersion

Measures of polarization for 1D opinion distributions

1. Spread
2. Dispersion
3. Coverage
4. Regionalization
5. Community fragmentation
6. Distinctness
7. Group divergence
8. Group consensus
9. **Size parity**



$$\text{size parity}_{\text{entropy}} = -\frac{1}{\ln G} \sum_{i=1}^G p_i \ln p_i.$$

Bramson, A. et al. *The Journal of Mathematical Sociology* **40**, 80–111 (2016).

Measures of polarization for 1D opinion distributions

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For any two senses one can construct an example from two distributions showing that **an increase in one sense of polarization does not (always) require or imply an increase** (and could be matched with a decrease) in the other sense.

Bramson, A. *et al.* *The Journal of Mathematical Sociology* **40**, 80–111 (2016).

Measures of polarization for 1D opinion distributions

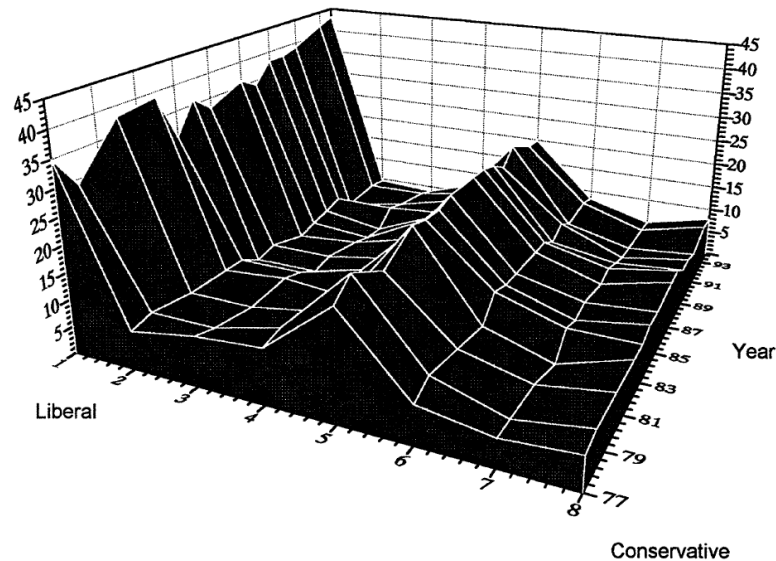
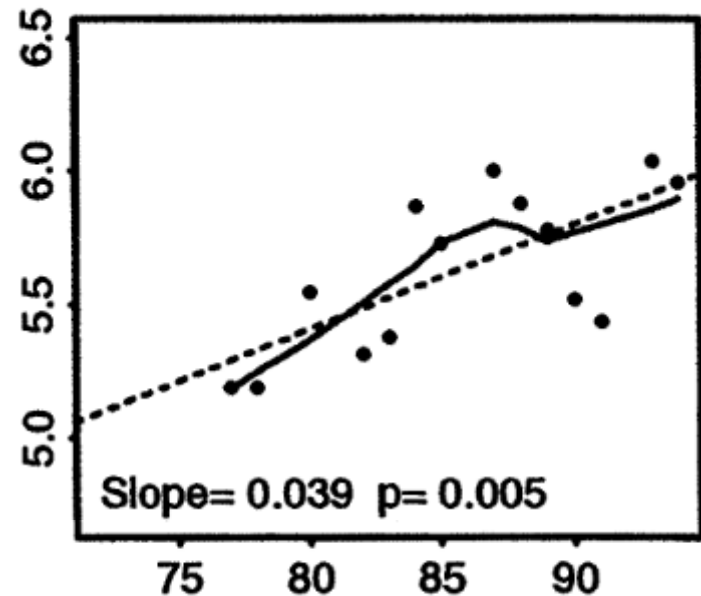


FIG. 4.—Distributions by year, attitudes toward abortion, full sample, GSS 1977–94.

Measure of dispersion

Variance



DiMaggio et al. *American Journal of Sociology* **102**, 690–755 (1996).

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Multidimensional
aspects (correlation)

DiMaggio et al. *American Journal of Sociology* **102**, 690–755 (1996).

Polarization of multidimensional attitudes: Correlation

Published in final edited form as:

AJS. 2008 January 28; 114(2): 408–446. doi:10.2139/ssrn.1010098.

Partisans without Constraint: Political Polarization and Trends in American Public Opinion¹

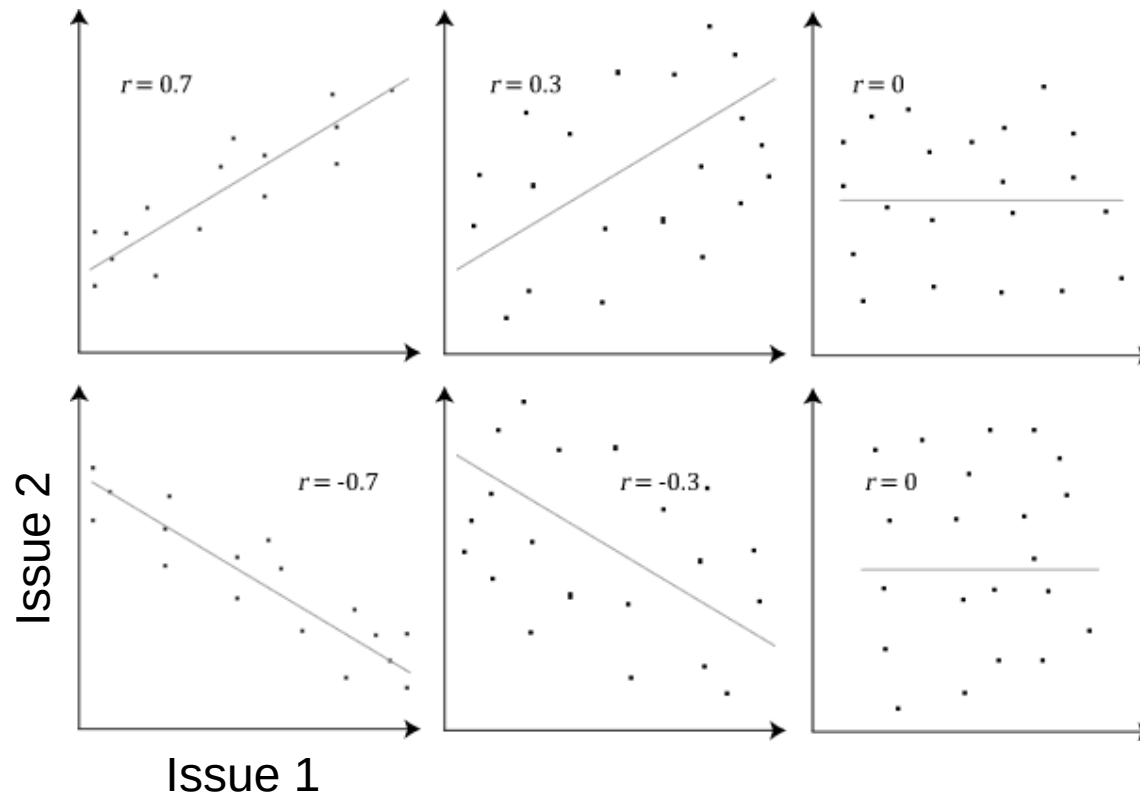
Delia Baldassarri and
Princeton University

Andrew Gelman
Columbia University

Abstract

Public opinion polarization is here conceived as a process of alignment along multiple lines of potential disagreement and measured as growing constraint in individuals' preferences. Using NES data from 1972 to 2004, the authors model trends in issue partisanship—the correlation of issue attitudes with party identification—and issue alignment—the correlation between pairs of issues—and find a substantive increase in issue partisanship, but little evidence of issue alignment.

Polarization of multidimensional attitudes: Correlation



Polarization of multidimensional attitudes: Correlation

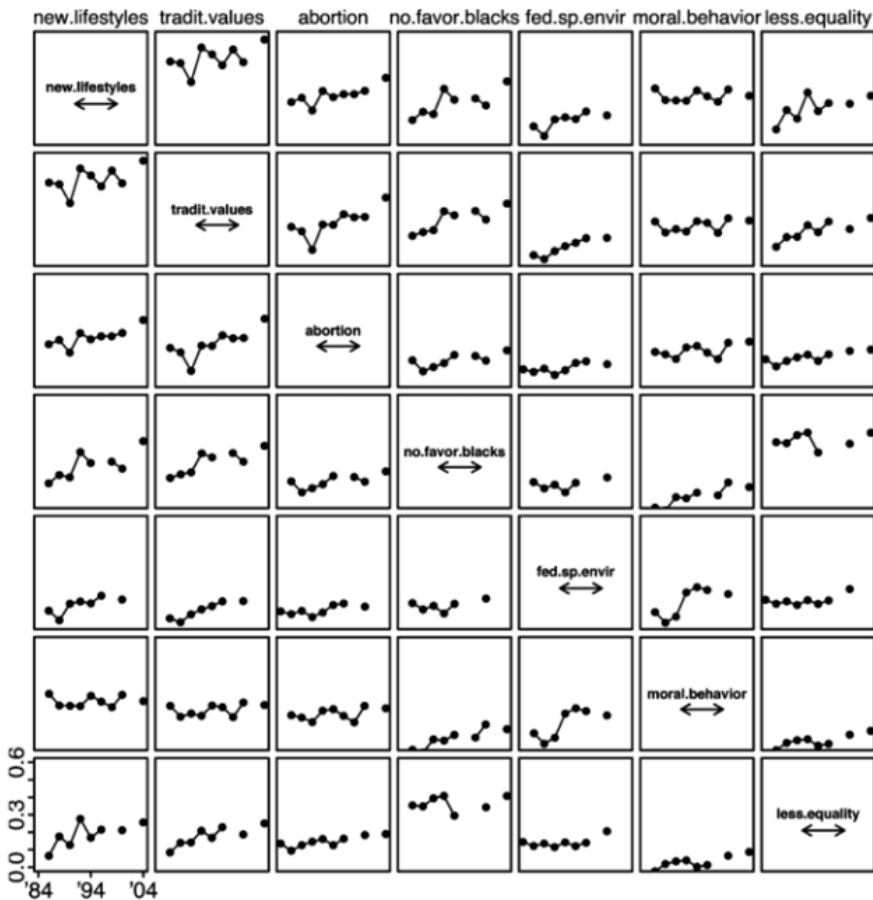


Fig. 6.

Trends in correlations between pairs of hot issues. The x -axis is time (1984–2004), and the y -axis is correlation (0 to .6). The plots are redundant: each pair of issues is plotted twice so that the reader can see on the same row (or column) the correlation between one issue and all the others.

Baldassarri et al. *AJS* **114**,
408–446 (2008).

Polarization of multidimensional attitudes: Correlation

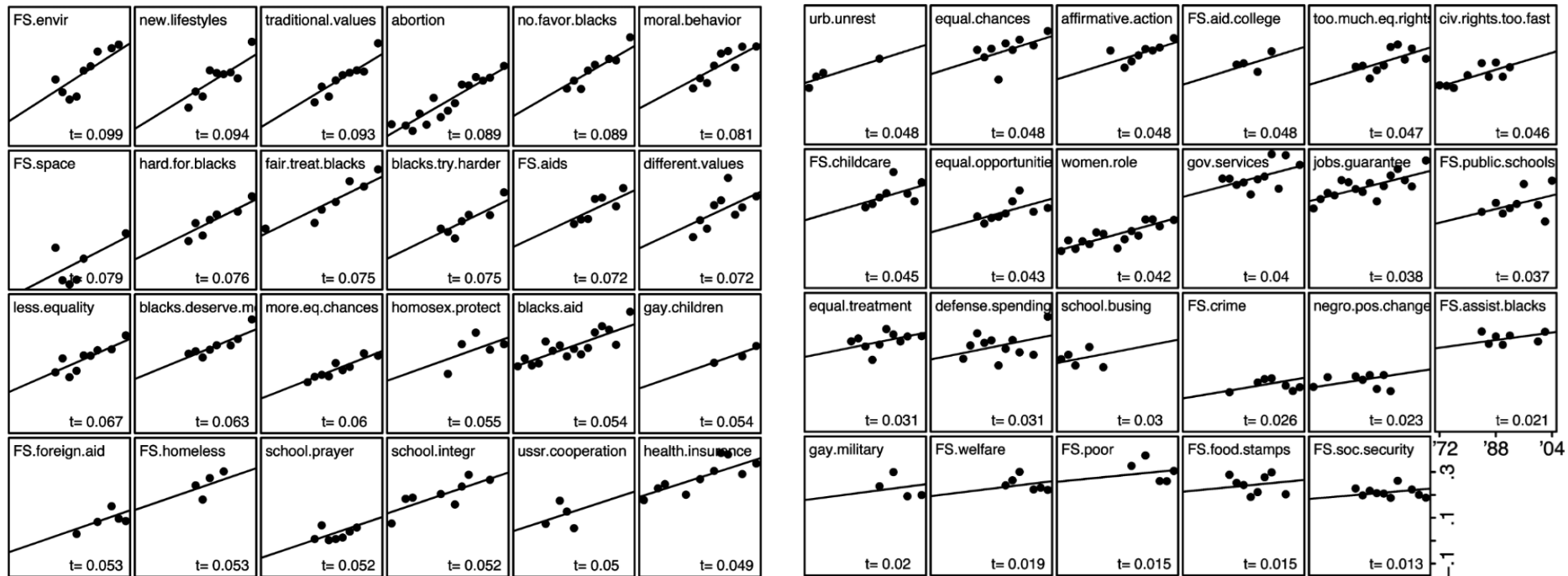


Fig. 3.

Trends in the correlation between issues and party identification. Regression lines as estimated in equation (1); at the bottom of each plot is reported the coefficient t_i . The x -axis is time (1972–2004), and the y -axis is correlation (–.1 to .4).

Baldassarri et al. *AJS* **114**,
408–446 (2008).

Polarization of multidimensional attitudes: Correlation

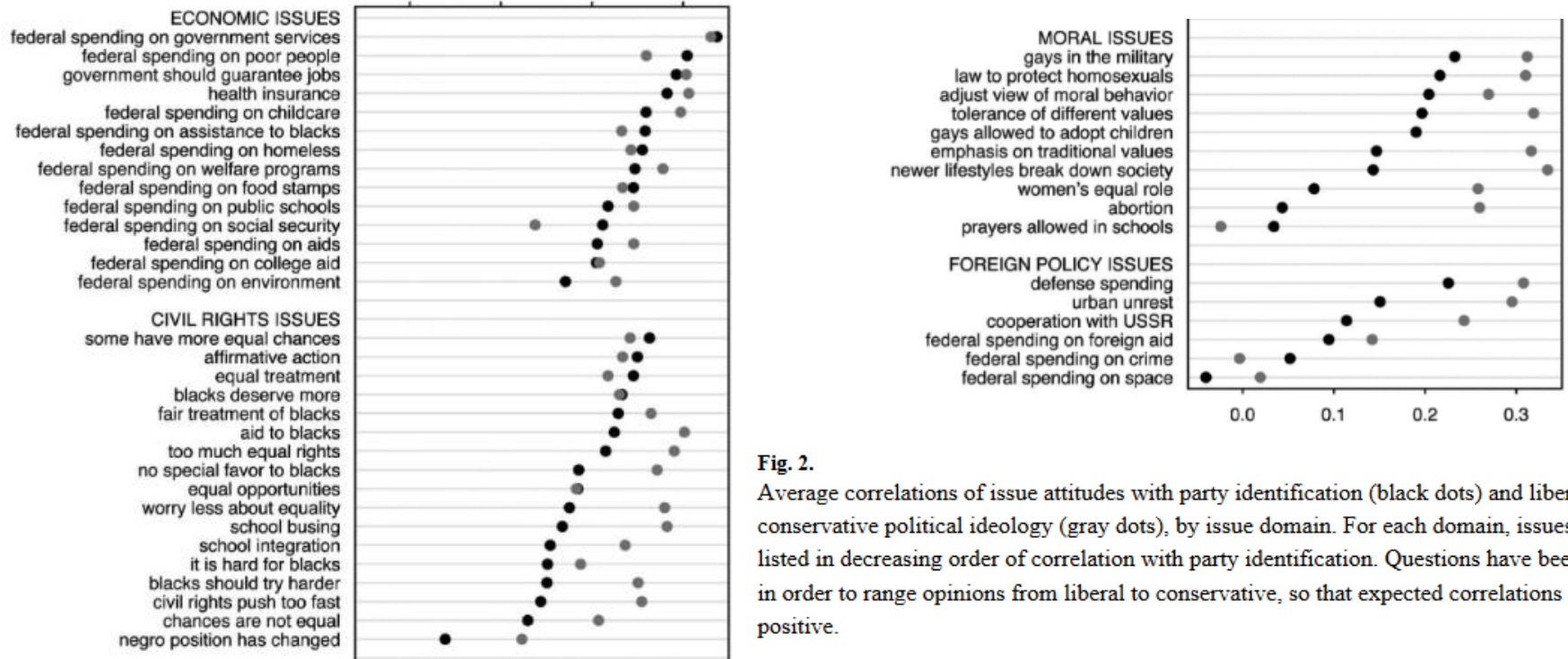


Fig. 2.

Average correlations of issue attitudes with party identification (black dots) and liberal-conservative political ideology (gray dots), by issue domain. For each domain, issues are listed in decreasing order of correlation with party identification. Questions have been coded in order to range opinions from liberal to conservative, so that expected correlations are all positive.

Baldassarri et al. *AJS* **114**,
408–446 (2008).



Designed by pikisuperstar / Freepik

Affective polarization

Affective polarization

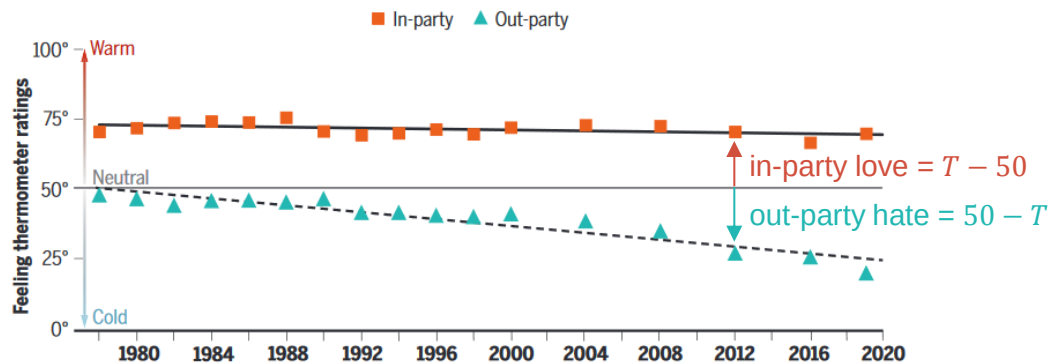
How to measure it?

- Survey self-reports of partisan affect.
- Implicit or subconscious tests of partisan bias.
- Behavioral measures of interpersonal trust and group favoritism or discrimination based on partisan cues.

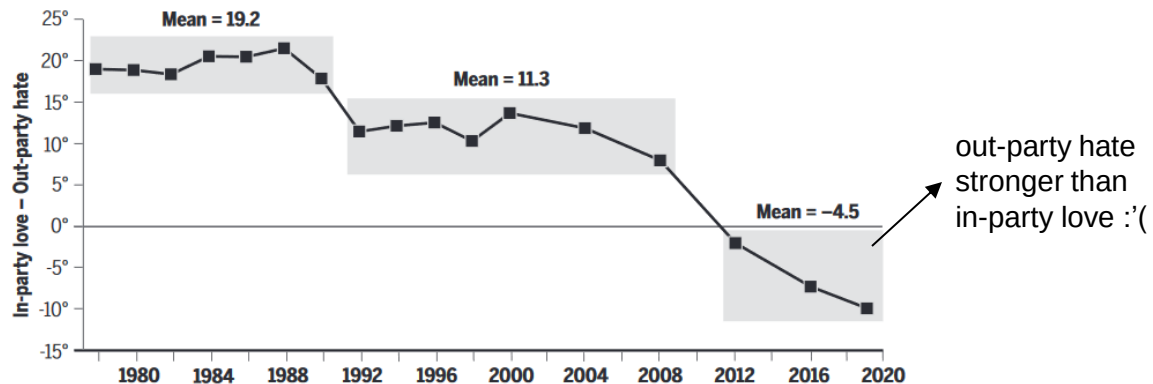
Iyengar et al. *Annu. Rev. Polit. Sci.* **22**, 129–146 (2019).

Affective polarization: surveys

Warmth toward the opposing party (out-party) has diminished for decades



Out-party hate has emerged as a stronger force than in-party love



Finkel, E. J. *et al. Science*
370, 533–536 (2020).

Affective polarization: implicit association

Implicit Association Test

Next, you will use the 'E' and 'I' computer keys to categorize items into groups as fast as you can. These are the four groups and the items that belong to each:

Category	Items
Good	Delightful, Joyous, Excitement, Fabulous, Triumph, Cheer, Fantastic, Spectacular
Bad	Hurtful, Sadness, Negative, Gross, Disaster, Tragic, Poison, Yucky
Joe Biden	
Barack Obama	

There are seven parts. The instructions change for each part. Pay attention!

Continue

Affective polarization: implicit association

Press "E" for
Joe Biden

Press "I" for
Barack Obama

Part 1 of 7

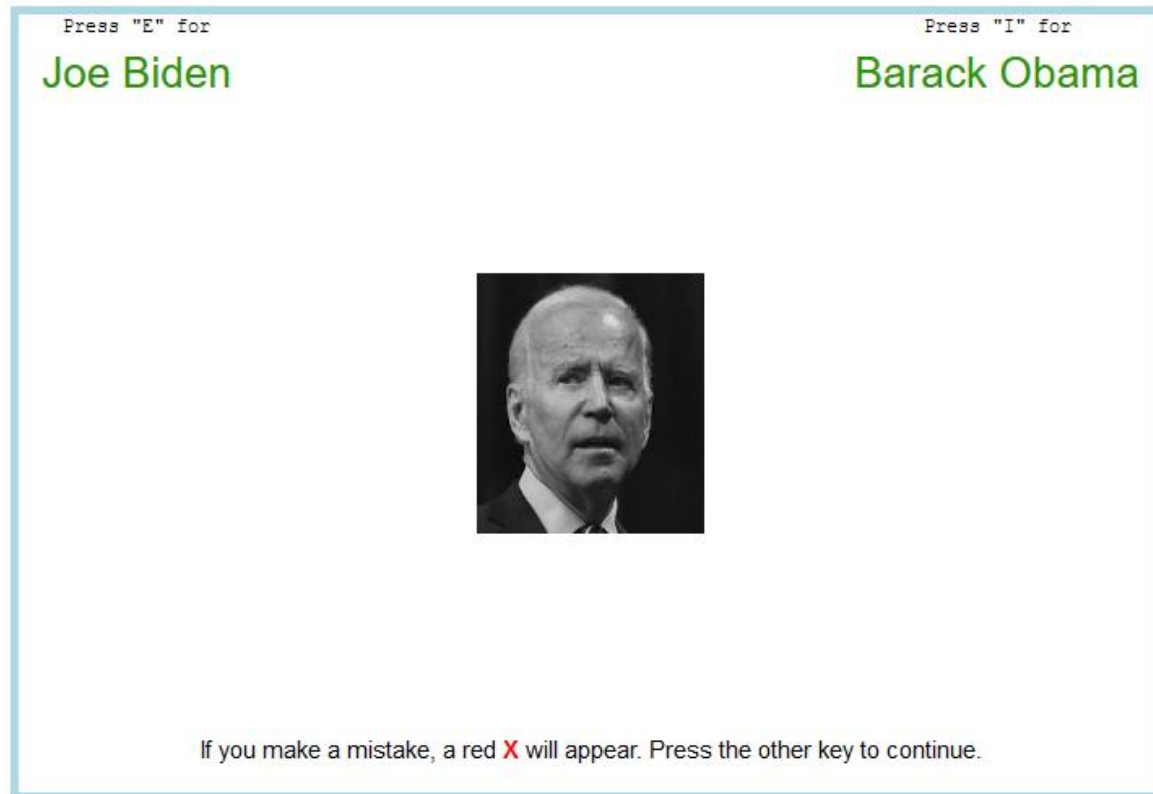
Put a left finger on the **E** key for items that belong to the category Joe Biden.
Put a right finger on the **I** key for items that belong to the category Barack Obama.
Items will appear one at a time.

If you make a mistake, a red **X** will appear. Press the other key to continue.
Go as fast as you can while being accurate.

Press the **space bar** when you are ready to start.

Greenwald et al. *Journal of Personality and Social Psychology* 74, 1464–1480 (1998).

Affective polarization: implicit association



Greenwald et al. *Journal of Personality and Social Psychology* **74**, 1464–1480 (1998).

Affective polarization: implicit association



Greenwald et al. *Journal of Personality and Social Psychology* **74**, 1464–1480 (1998).

Affective polarization: implicit association

Press "E" for

Bad

Press "I" for

Good

Part 2 of 7

Put a left finger on the **E** key for items that belong to the category **Bad**.
Put a right finger on the **I** key for items that belong to the category **Good**.

If you make a mistake, a red **X** will appear. Press the other key to continue.
Go as fast as you can while being accurate.

Press the **space bar** when you are ready to start.

Greenwald et al. *Journal of Personality and Social Psychology* **74**, 1464–1480 (1998).

Affective polarization: implicit association



Greenwald et al. *Journal of Personality and Social Psychology* **74**, 1464–1480 (1998).

Affective polarization: implicit association

Press "E" for

Bad

or

Joe Biden

Press "I" for

Good

or

Barack Obama

Part 3 of 7

Use the **E** key for **Joe Biden** and for **Bad**.
Use the **I** key for **Barack Obama** and for **Good**.
Each item belongs to only one category.

If you make a mistake, a red **X** will appear. Press the other key to continue.
Go as fast as you can while being accurate.

Press the **space bar** when you are ready to start.

Greenwald et al. *Journal of Personality and Social Psychology* **74**, 1464–1480 (1998).

Affective polarization: implicit association



Greenwald et al. *Journal of Personality and Social Psychology* **74**, 1464–1480 (1998).

Affective polarization: implicit association



Greenwald et al. *Journal of Personality and Social Psychology* **74**, 1464–1480 (1998).

Affective polarization: implicit association

Press "E" for
Bad
or
Joe Biden

Press "I" for
Good
or
Barack Obama

Part 4 of 7

This is the same as the previous part.
Use the **E** key for Joe Biden and for Bad.
Use the **I** key for Barack Obama and for Good.
Each item belongs to only one category.

Go as fast as you can while being accurate.

Press the **space bar** when you are ready to start.

Press "E" for
Barack Obama

Press "I" for
Joe Biden

Part 5 of 7

Watch out, the labels have changed position!
Use the left finger on the **E** key for Barack Obama.
Use the right finger on the **I** key for Joe Biden.

Go as fast as you can while being accurate.

Press the **space bar** when you are ready to start.

Press "E" for
Bad
or
Barack Obama

Press "I" for
Good
or
Joe Biden

Part 6 of 7

Use the **E** key for Barack Obama and for Bad.
Use the **I** key for Joe Biden and for Good.
Each item belongs to only one category.

If you make a mistake, a red **X** will appear. Press the other key to continue.
Go as fast as you can while being accurate.

Press the **space bar** when you are ready to start.

Press "E" for
Bad
or
Barack Obama

Press "I" for
Good
or
Joe Biden

Part 7 of 7

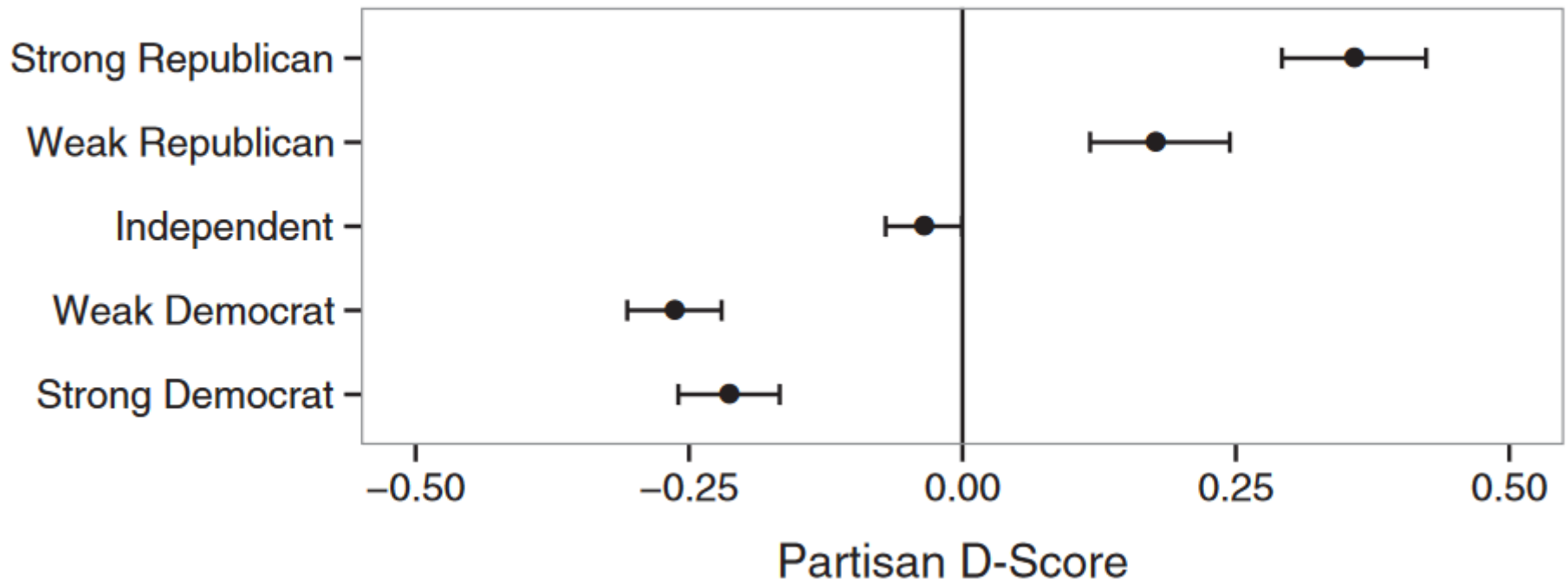
This is the same as the previous part.
Use the **E** key for Barack Obama and for Bad.
Use the **I** key for Joe Biden and for Good.
Each item belongs to only one category.

Go as fast as you can while being accurate.

Press the **space bar** when you are ready to start.

Affective polarization: implicit association

Implicit Partisan Affect by Partisan Strength



D-score: Normalized latence difference between positive associations to each group.

Iyengar et al. *American Journal of Political Science* **59**, 690–707 (2015).

Affective polarization: behavioral measures (games)

Trust game (2 player *game*):

- Player 1 gets 10\$.
- Decides how much to give to Player 2.
- Player 2 receives 3x that amount and decides how much to give back to Player 1.

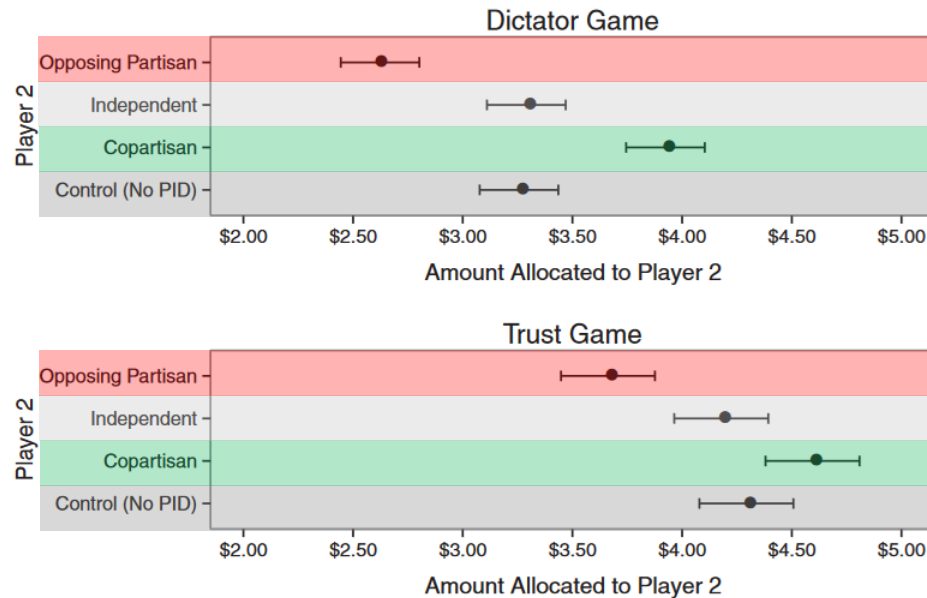
Dictator game (2 player *game*):

- Player 1 gets 10\$.
- Decides how to split it with Player 2.

Iyengar et al. *American Journal of Political Science* **59**, 690–707 (2015).

Affective polarization: behavioral measures (games)

FIGURE 9 Effects of Ingroup and Outgroup Membership on Allocations to Player 2



Note: The dots are the means for each group, and the bars are 95% confidence intervals.

Iyengar et al. *American Journal of Political Science* **59**, 690–707 (2015).

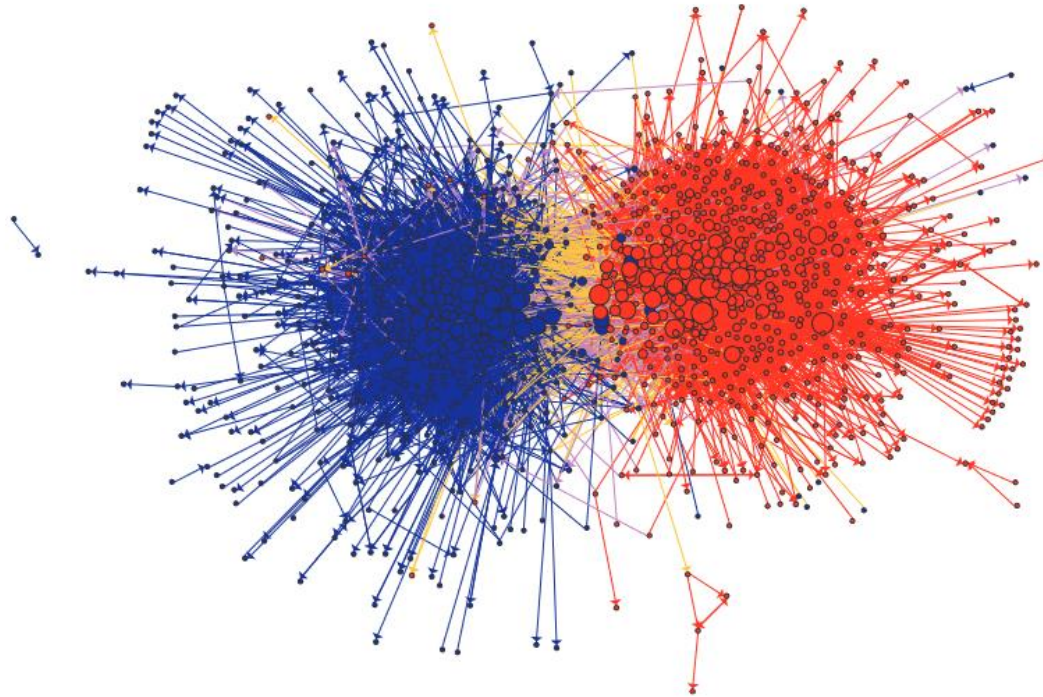


Figure 1: Community structure of political blogs (expanded set), shown using utilizing the GUESS visualization and analysis tool[2]. The colors reflect political orientation, red for conservative, and blue for liberal. Orange links go from liberal to conservative, and purple ones from conservative to liberal. The size of each blog reflects the number of other blogs that link to it.

Network polarization

Adamic et al. in *Proceedings of the 3rd international workshop on Link discovery*
36–43 (Association for Computing
Machinery, 2005)

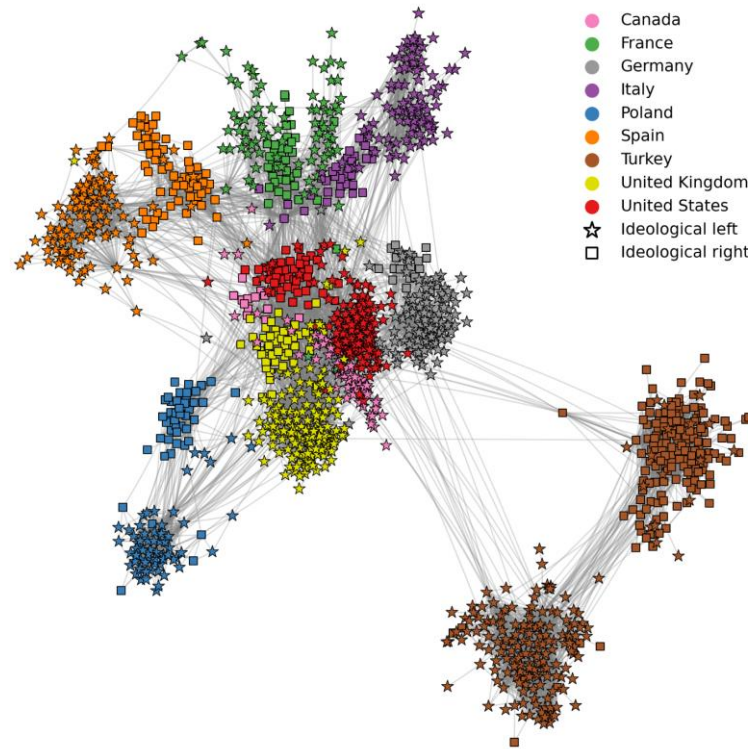


FIG. 1. **The network of political Twitter interactions is segregated across countries and polarized within them.** The figure depicts a co-retweet network where nodes correspond to individual politicians and an edge is drawn between two nodes if those politicians share a common retweeter (see Methods). Square nodes correspond to the ideological right, star nodes correspond to the ideological left, as determined using the latent ideology (see Fig. 2). The network visualization is produced using a force directed drawing algorithm (see Methods).

Network polarization

Falkenberg et al. “Affective and interactional polarization align across countries”, *arXiv:2311.18535*

Network polarization: quantifying controversy with network communities

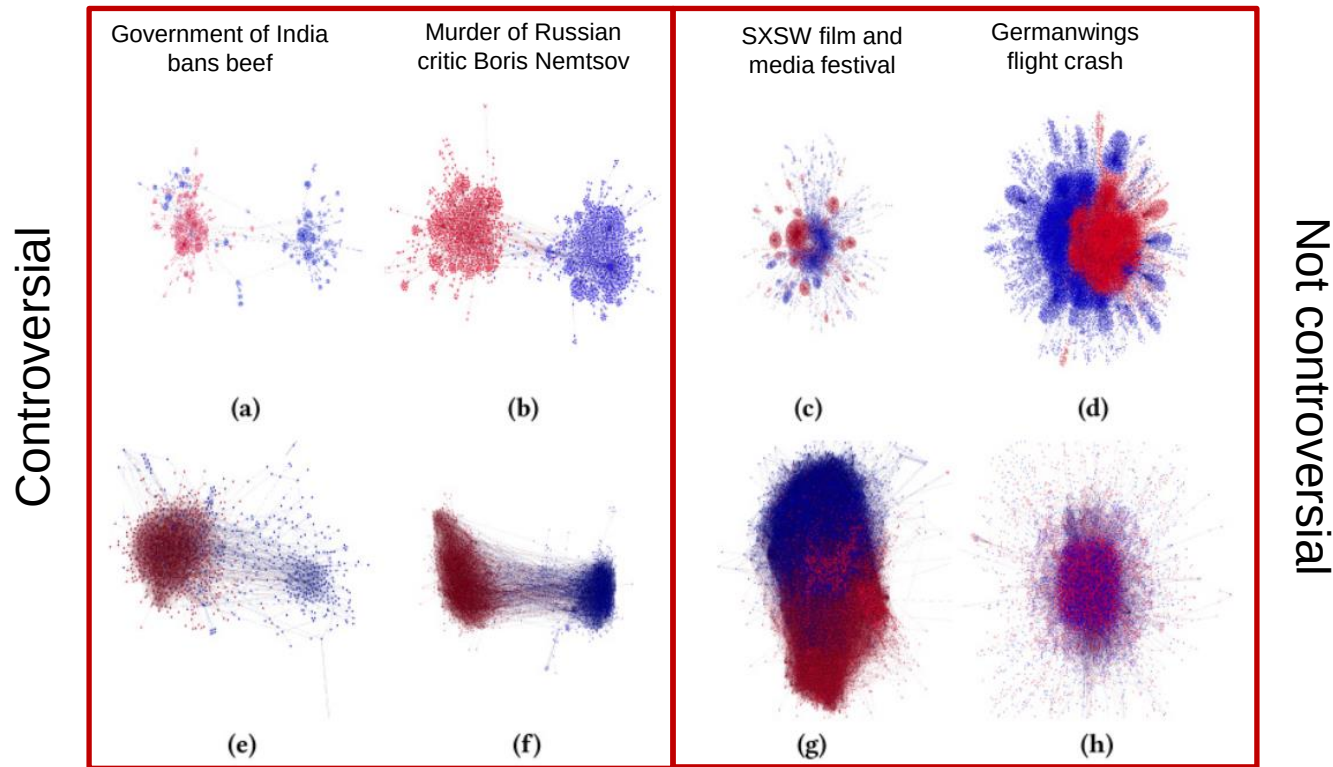


Fig. 3. Sample conversation graphs with retweet (top) and follow (bottom) aspects (visualized using the force-directed layout algorithm in Gephi). The left side is controversial, ((a) and (e)) #beefban and ((b) and (f)) #russia_march, while the right side is non-controversial, ((c) and (g)) #sxsw and ((d) and (h)) #germanwings. Only the largest connected component is shown.

Garimella et al., *Trans. Soc. Comput.* **1**, 3:1-3:27 (2018).

Network polarization: quantifying controversy with network communities

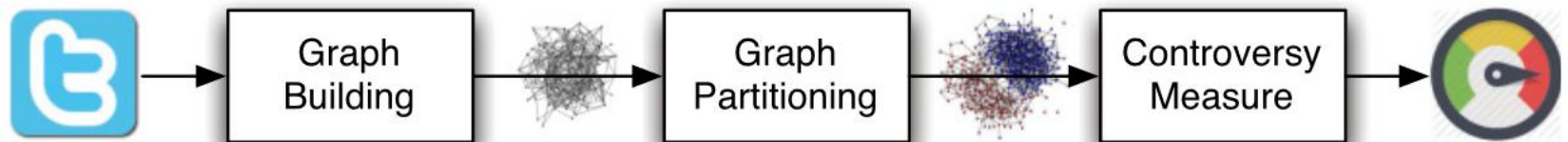


Fig. 1. Block diagram of the pipeline for computing controversy scores.

Garimella et al., *Trans. Soc. Comput.* **1**, 3:1-3:27 (2018).

Network polarization: quantifying controversy with network communities

- **BCC**: Compare bridge degree of the links across communities with the links of the rest of the nw.
- **EC**: 1) Embed nodes in 2D space. 2) Compare within-group and between-groups node distances.
- **GMKC**: 1) Identify boundary of each community. 2) Compare connectivity of boundary nodes to internal nodes vs other boundary nodes.
- **MBLB**: 1) Infer 1D opinion distribution from nw. 2) Measure polarization combining size parity and group divergence.
- **RWC**: Probability for random walks starting in community 1 to cross to community 2 (random walk stops when it reaches high degree node).

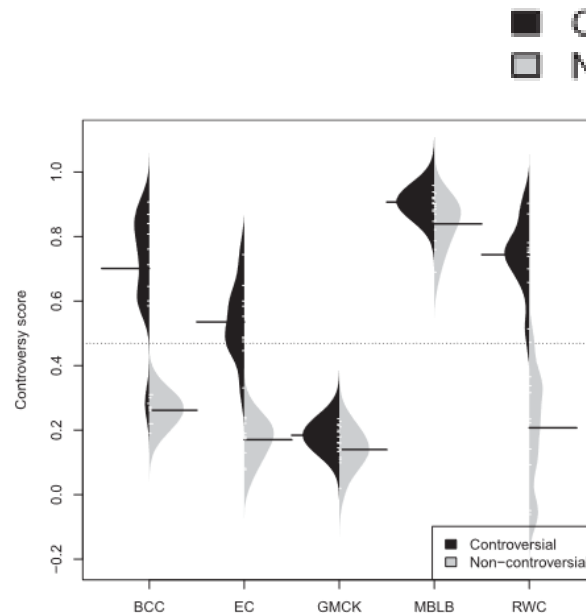


Fig. 7. Controversy scores on *retweet* graphs of various controversial and non-controversial datasets.

■ Controversial
■ Non-controversial

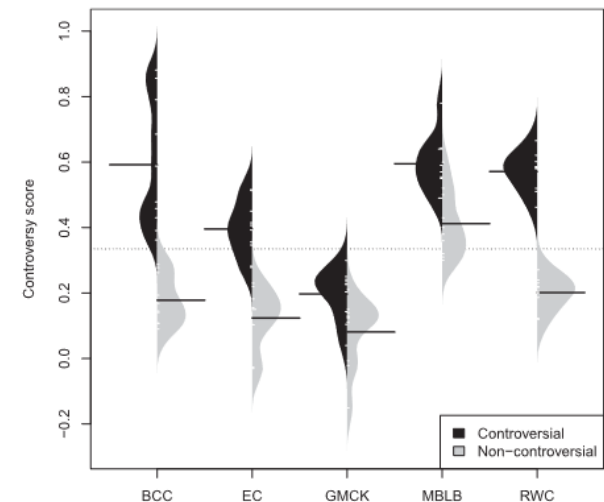
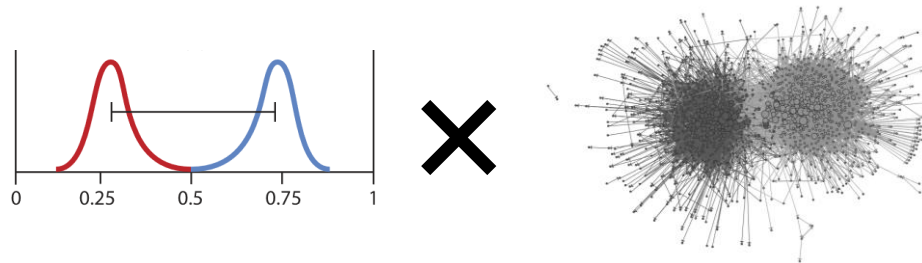


Fig. 8. Controversy scores on *follow* graphs of various controversial and non-controversial datasets.

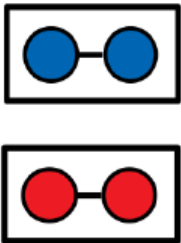
Garimella et al., *Trans. Soc. Comput.* **1**, 3:1-3:27 (2018).

Network polarization: homogeneous echo chambers

-Is the existence of community structure and/or a bimodal opinion distribution enough to say that there's polarization?



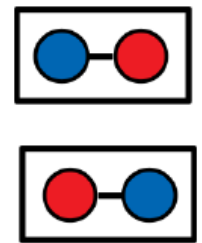
CONSENSUS



COEXISTENCE



POLARIZATION



A. F. Peralta et al. *Phys Rev E* 104, 044312 (2021)

A. F. Peralta et al. *J Phys Complex* 2, 045009 (2021)

Network polarization: homogeneous echo chambers

-Is the existence of community structure enough for polarization to emerge?

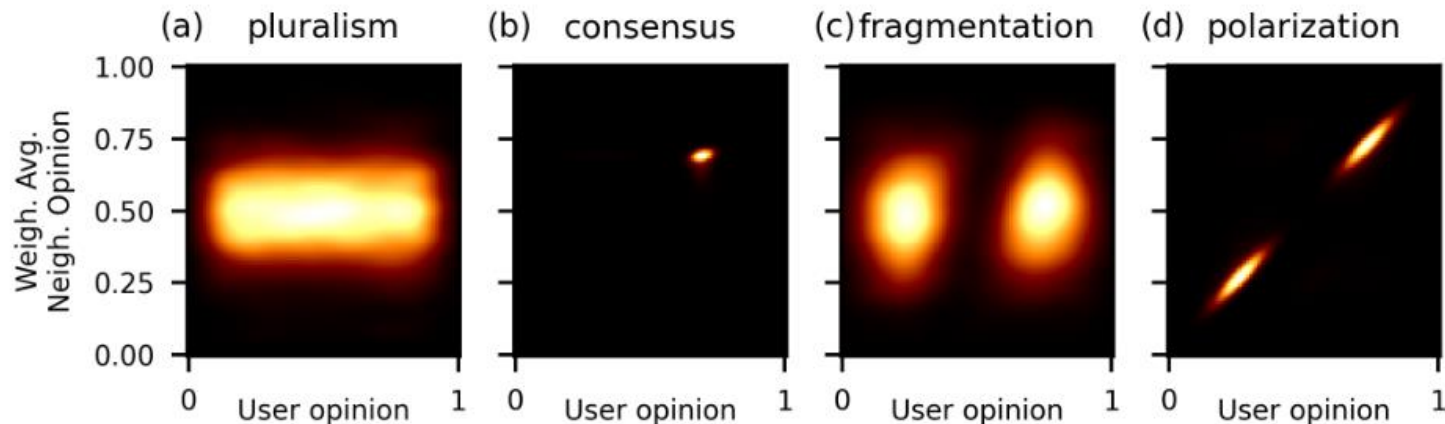
We can look at the **correlation** of the **opinion of a node** with the **average opinion of its neighbors**!

Valensise et al. Preprint at
<https://doi.org/10.48550/arXiv.2205.15958> (2022).

Network polarization: homogeneous echo chambers

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Network polarization: homogeneous echo chambers



Ideology: Links to news outlets with known political leaning.

Network: Followers.



Ideology: Links to news outlets with known political leaning.

Network: There exists a direct link from node i to node j if user i comments on a submission or comment by user j .



Ideology: Endorsements in the form of likes to posts from pages with known political leaning.

Network: Two users are connected if they cocommented on at least one post.



Ideology: Links to news outlets with known political leaning.

Network: Two users are connected if they cocommented on at least one post.

Cinelli, M. et al. *PNAS* **118**, e2023301118 (2021).

Network polarization: homogeneous echo chambers

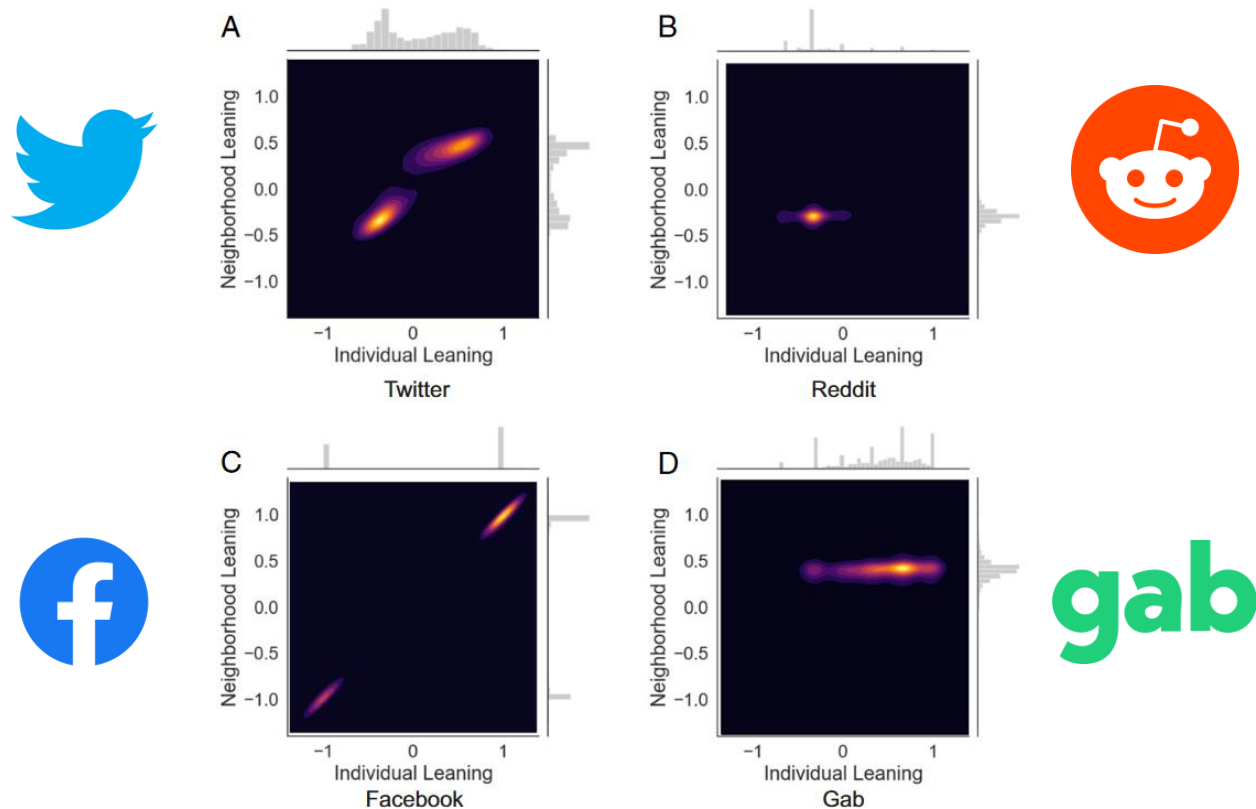


Fig. 1. Joint distribution of the leaning of users x and the average leaning of their neighborhood x^{NN} for different datasets. (A) Twitter, (B) Reddit, (C) Facebook, and (D) Gab. Colors represent the density of users: The lighter the color, the larger the number of users. Marginal distribution $P(x)$ and $P^N(x)$ are plotted on the x and y axes, respectively. Facebook and Twitter present by homophilic clustering.

Cinelli, M. et al. *PNAS* **118**, e2023301118 (2021).

Recap of polarization

Attitudinal polarization

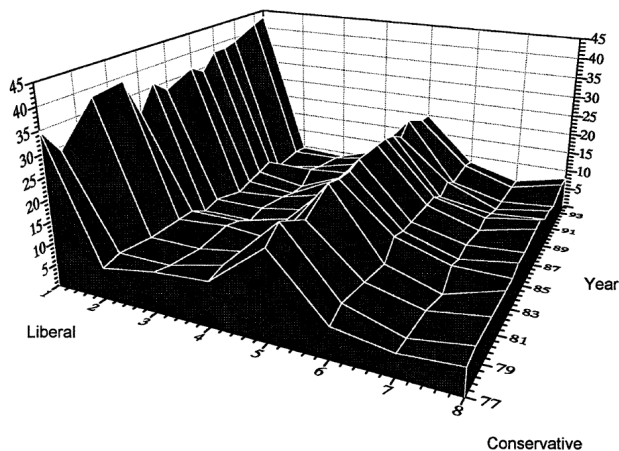
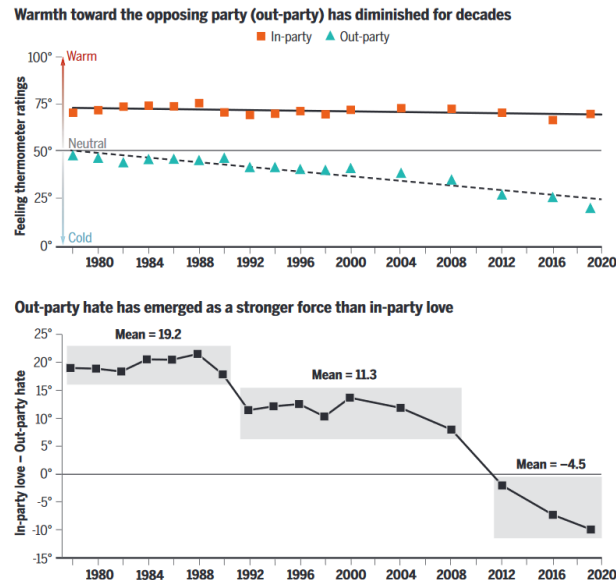


FIG. 4.—Distributions by year, attitudes toward abortion, full sample, GSS 1977–94.

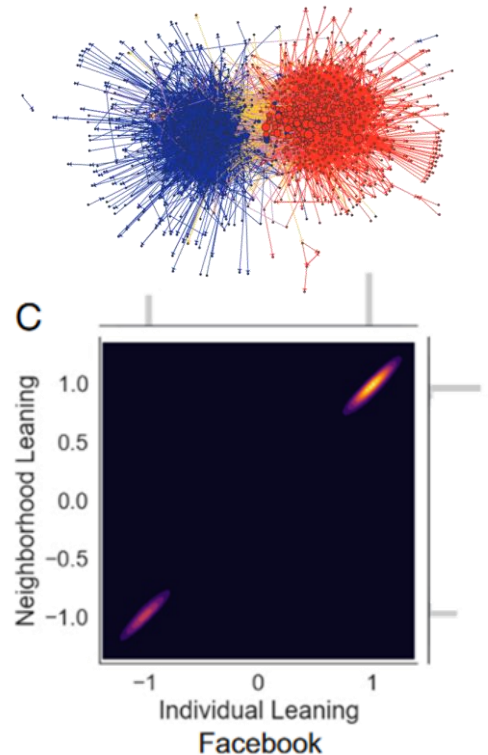
- One-dimensional measures (e.g. dispersion).
- Multidimensional measures (correlation).

Affective polarization



- Sentiment thermometer.
- Implicit association.
- Behavioral asymmetries (games).

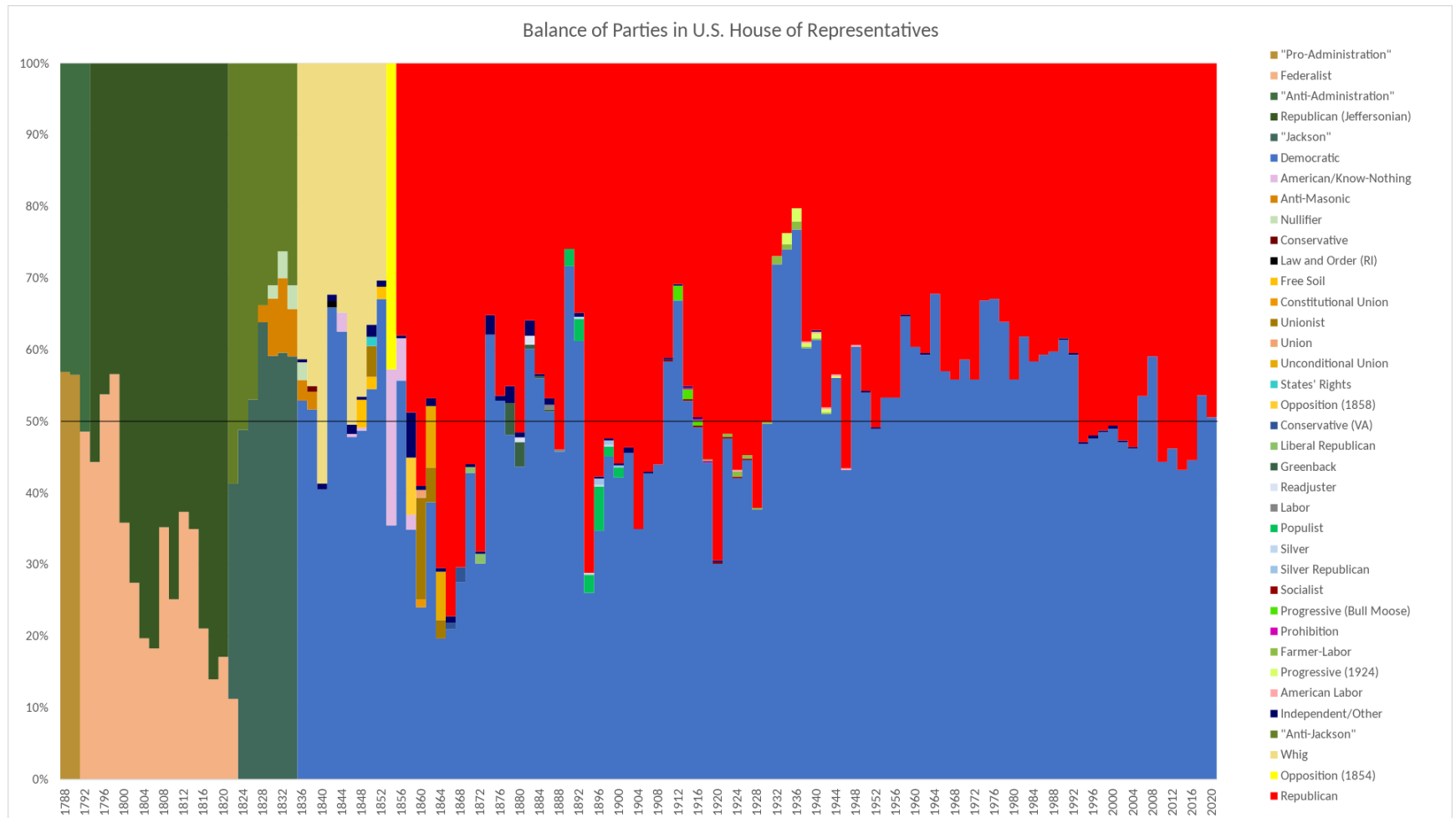
Network polarization



- Poor inter-community connections.
- Node's opinion similar to neighbors'.

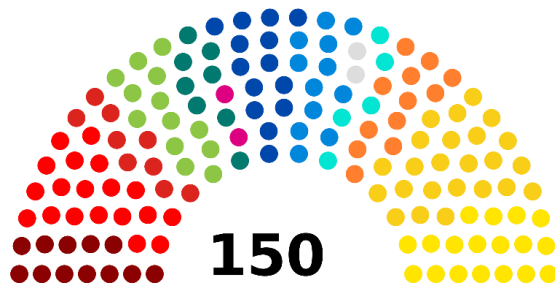
MULTI-POLARIZATION

Bi-polarization in the US

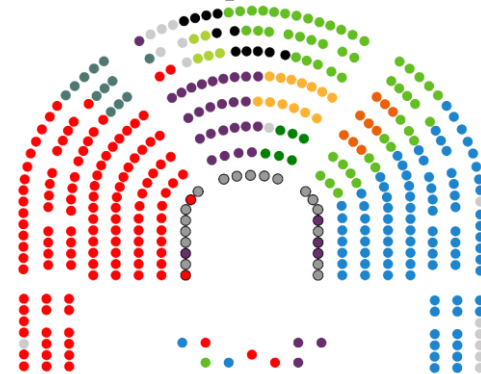


Multi-party democracies

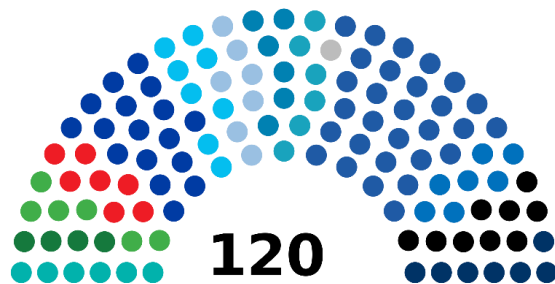
Belgium



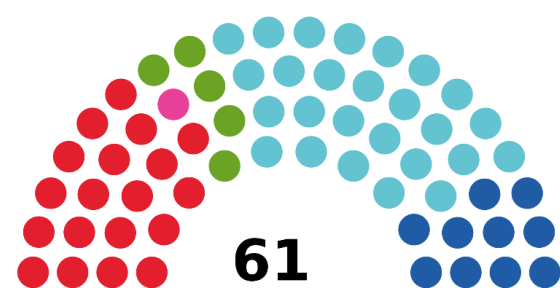
Spain



Israel



Austria

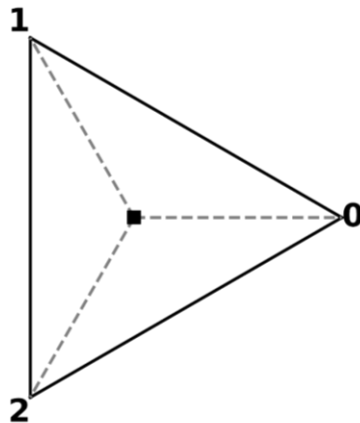


Multi-polarization: opinion space

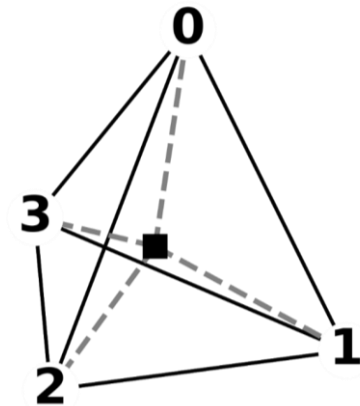
Bipolar space-1D



Tripolar space-2D

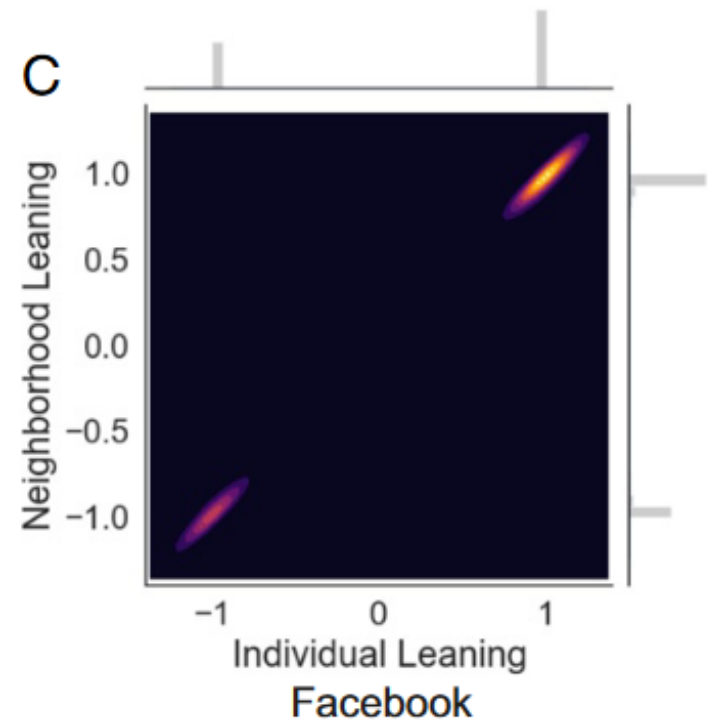
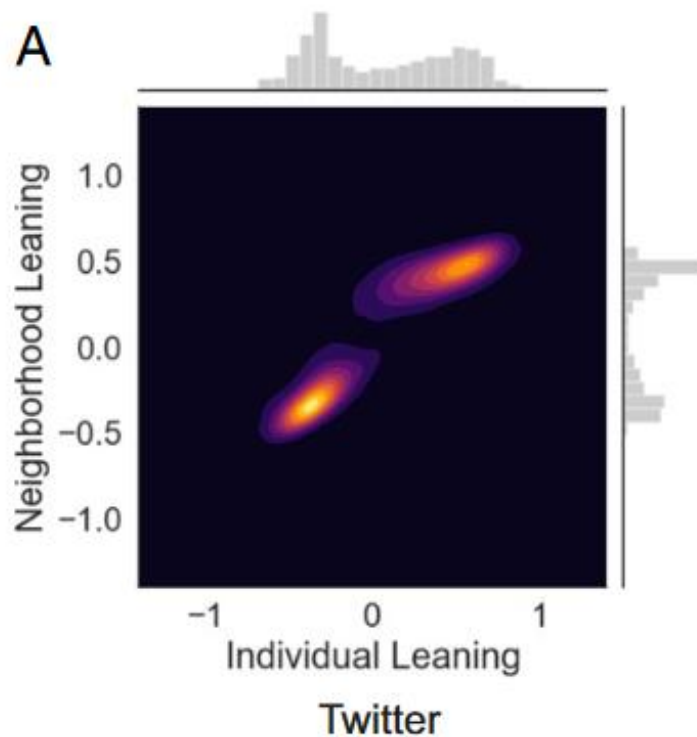


Quadripolar space-3D



Martin-Gutierrez et al. *Chaos, Solitons & Fractals* **169**, 113244 (2023).

Multi-polarization: opinion inference



Martin-Gutierrez et al. *Chaos, Solitons & Fractals* **169**, 113244 (2023).

Multi-polarization: opinion inference

Proceedings of the Ninth International AAAI Conference on Web and Social Media

What Do Retweets Indicate? Results from User Survey and Meta-Review of Research

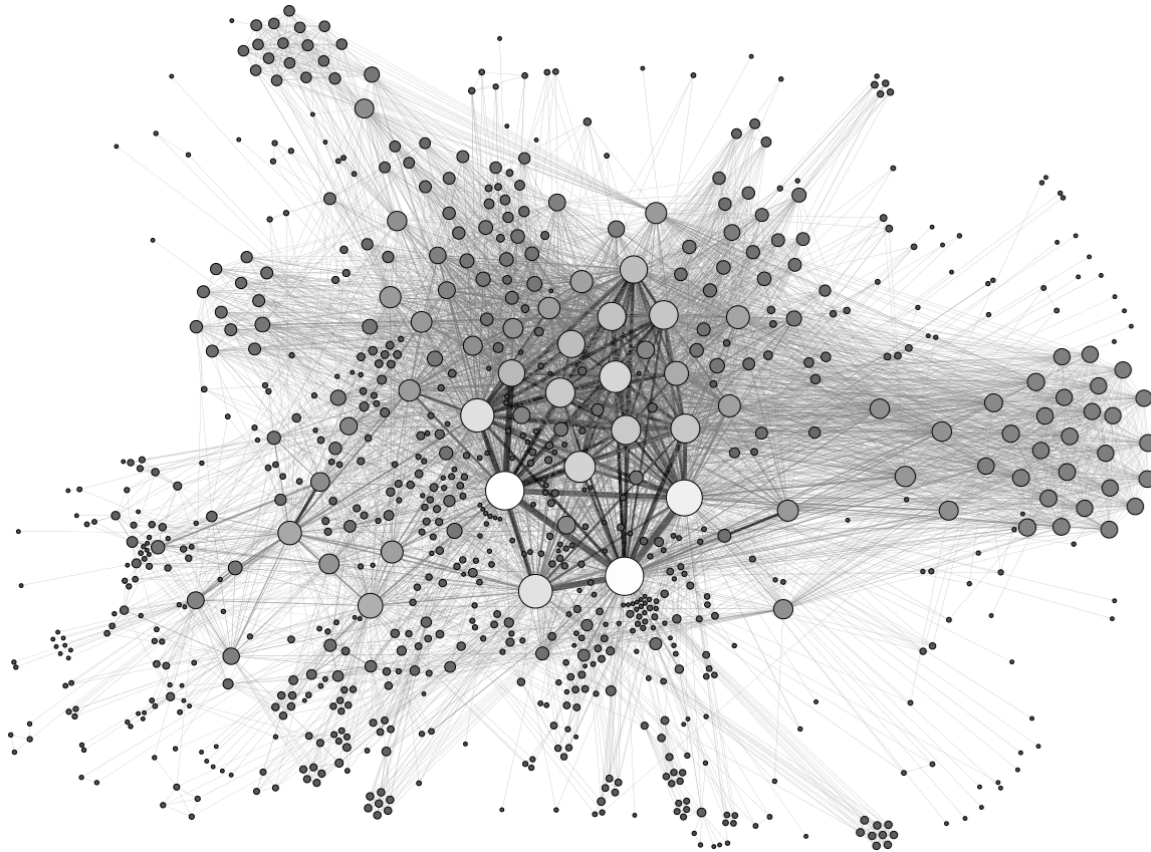
P. T. Metaxas, E. Mustafaraj, K. Wong, L. Zeng, M. O’Keefe, S. Finn

their significance. In this paper we answer the question of what users indicate when they retweet. We do so in a comprehensive fashion, by employing a user survey, a study of user profiles, and a meta-review of over 100 research publications from three related major conferences. Our findings indicate that retweeting indicates not only interest in a message, but also trust in the message and the originator, and agreement with the message contents. However, the findings are signif-

Martin-Gutierrez et al. *Chaos, Solitons & Fractals* **169**, 113244 (2023).

Multi-polarization: opinion inference

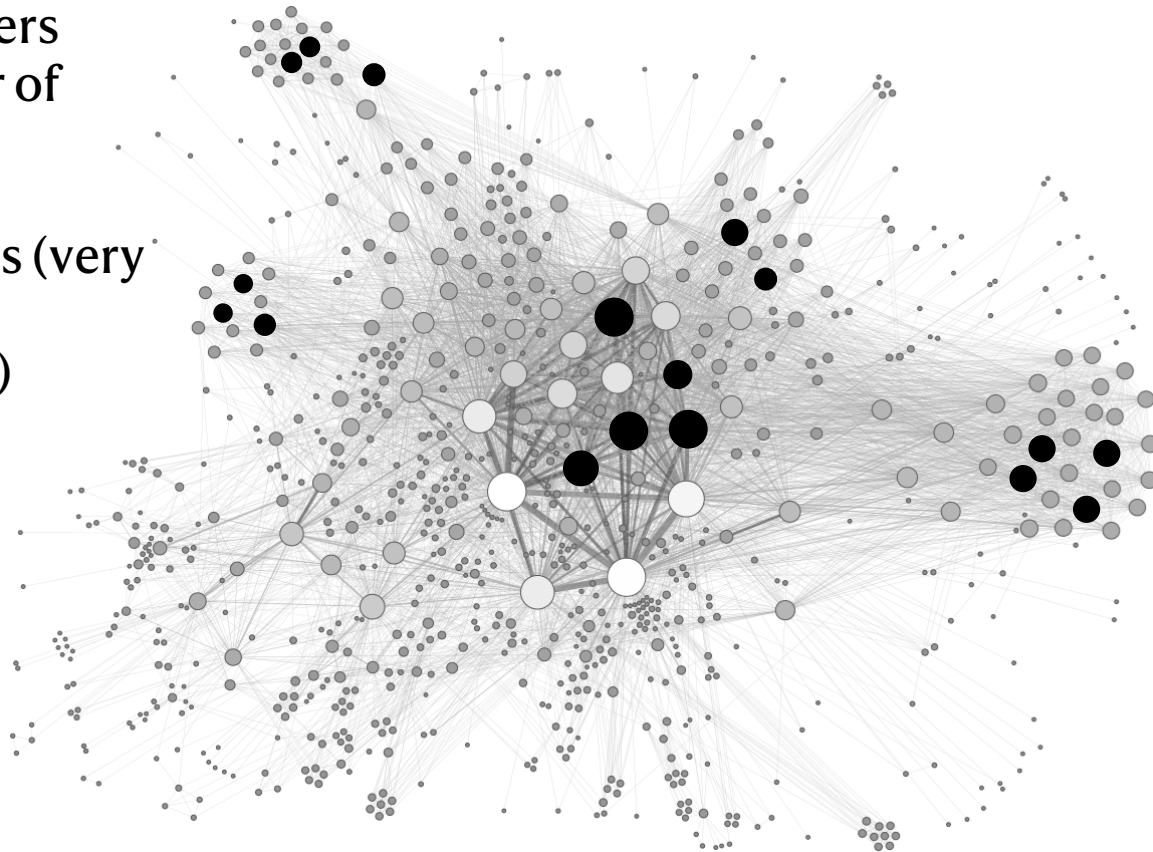
Retweet
network



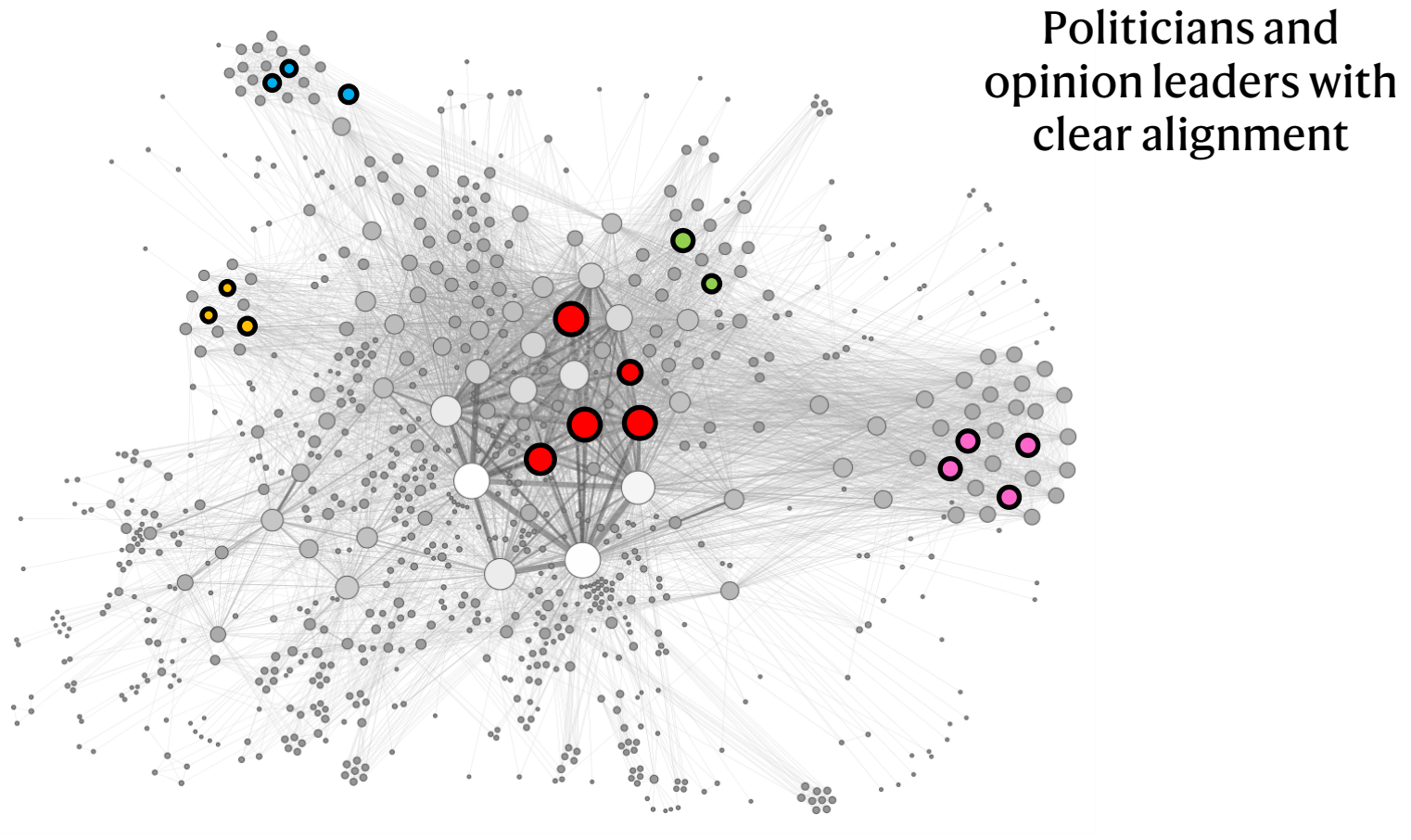
Multi-polarization: opinion inference

Influential users
(high number of
retweets)

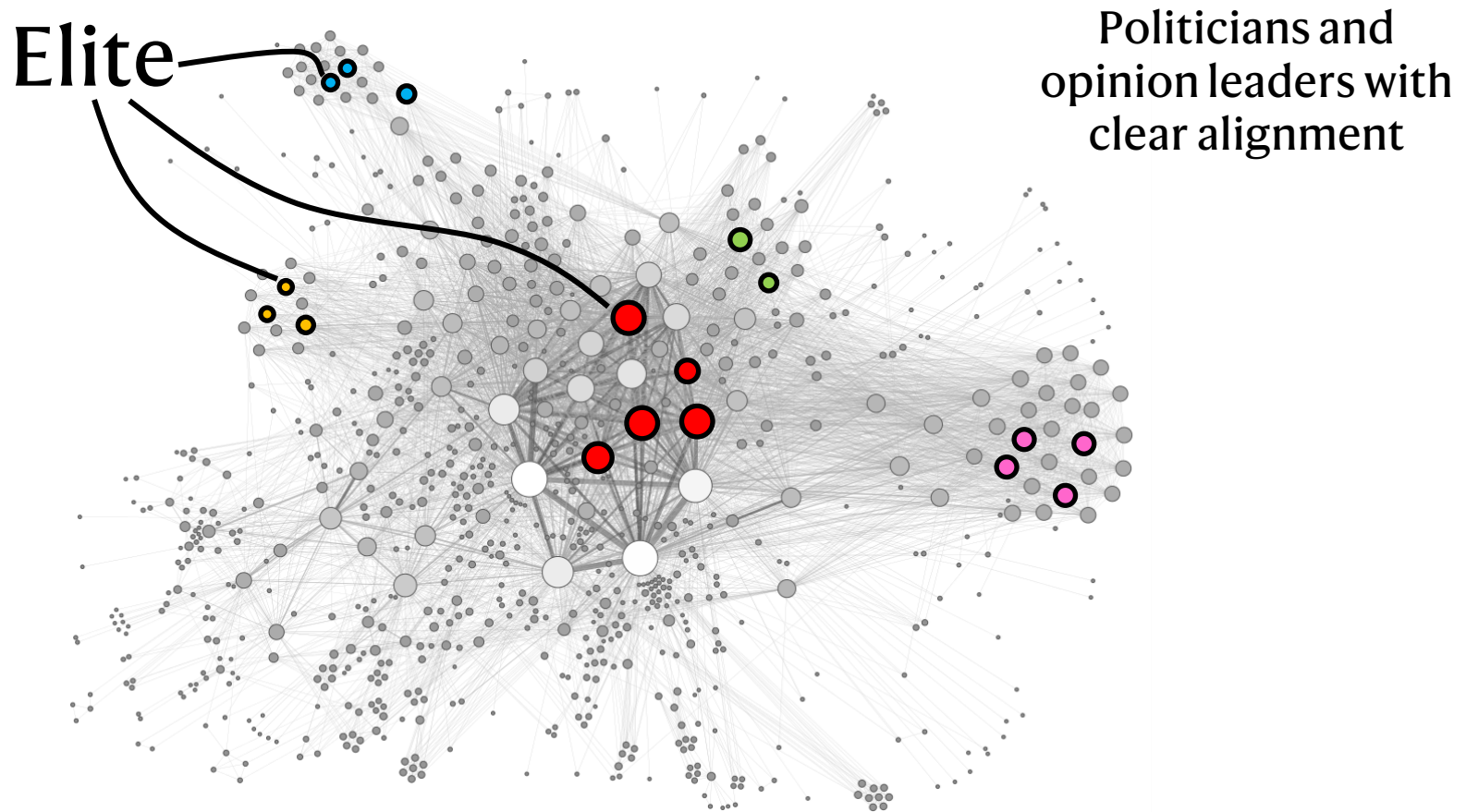
Engaged users (very
active in the
conversation)



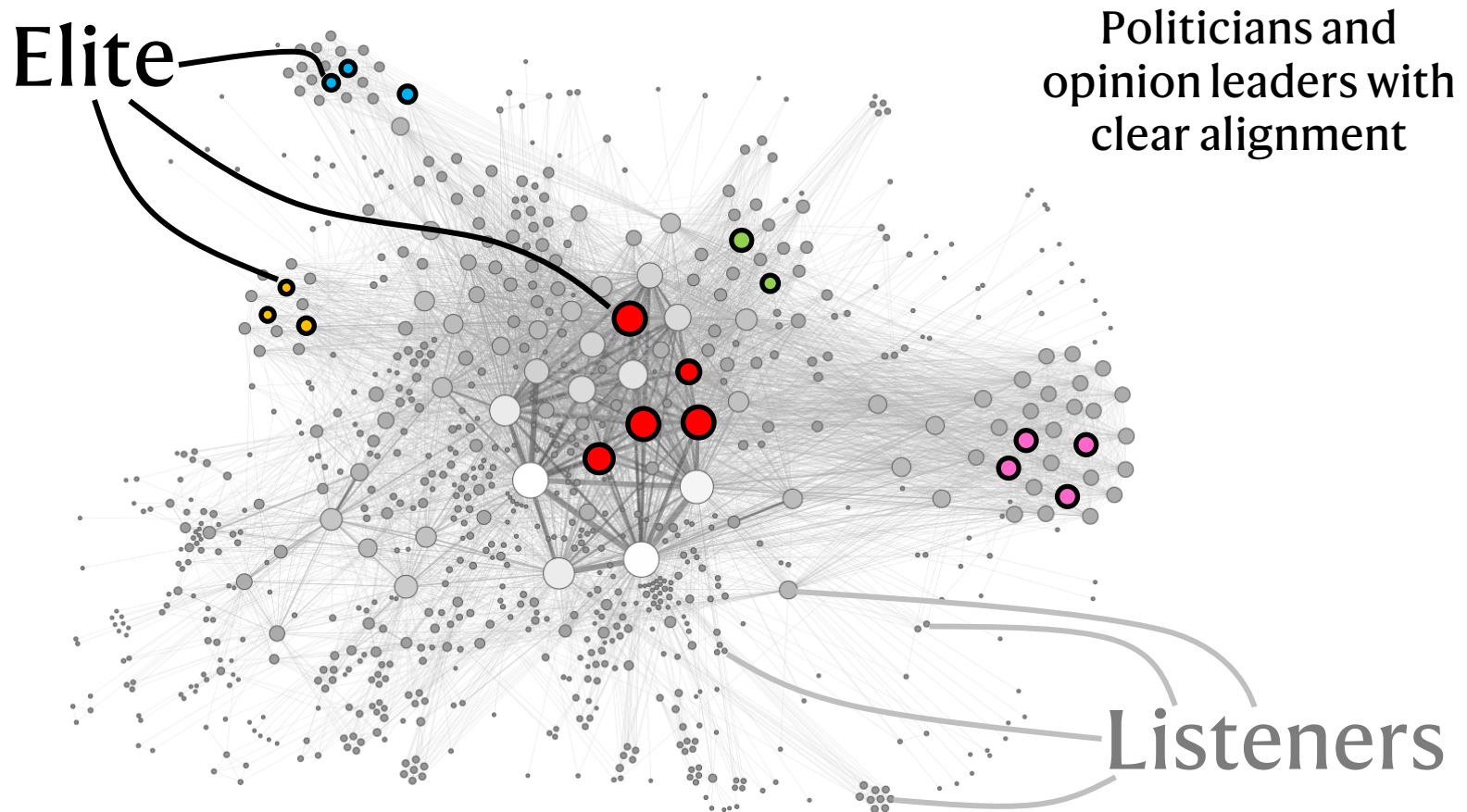
Multi-polarization: opinion inference



Multi-polarization: opinion inference



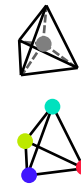
Multi-polarization: opinion inference



Multi-polarization: opinion inference

$\vec{x}_i \rightarrow$ Opinion vector of node i .

$$\vec{x}_i(t = 0) = \begin{cases} \vec{0} & \text{if } i \text{ is Listener} \\ \vec{x}_p & \text{if } i \text{ is Elite} \end{cases}$$

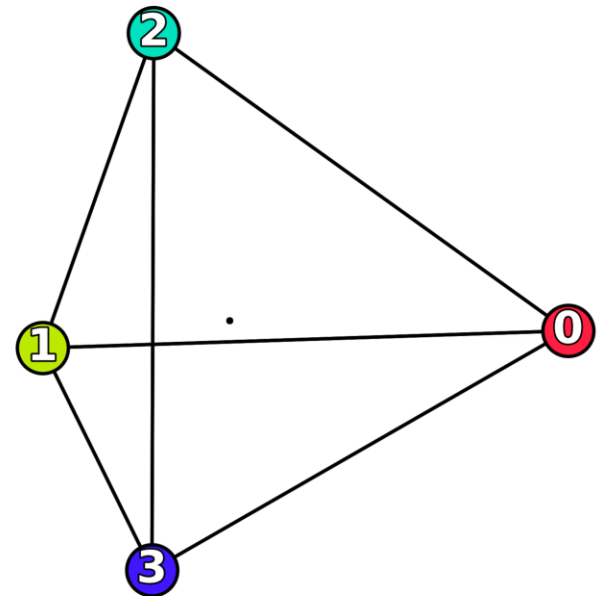
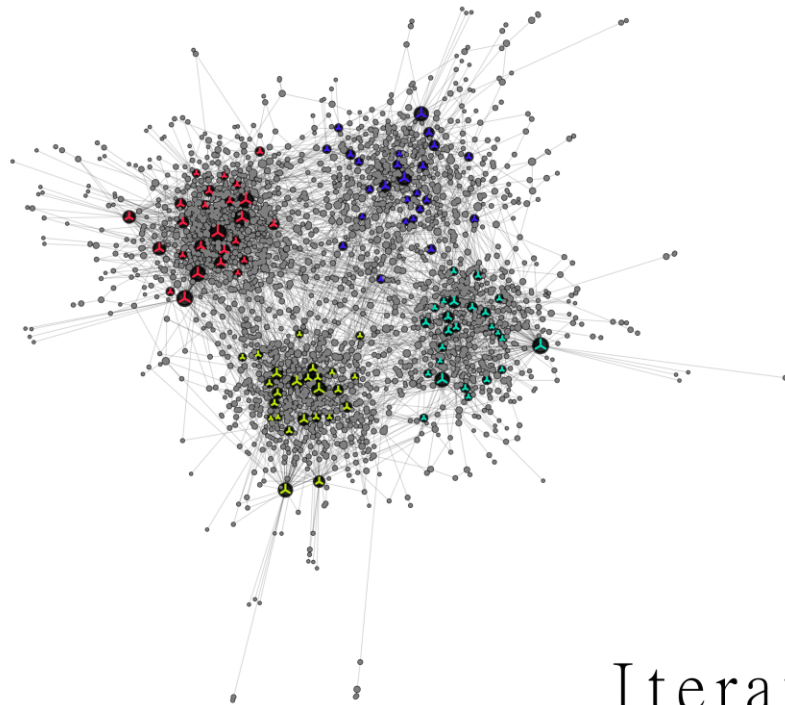


Iteratively average the opinion of the neighbors:

$$\vec{x}_i(t) = \frac{\sum_{j \in N(i)} \vec{x}_j(t-1)}{|N(i)|}$$

$N(i) \rightarrow$ Set of neighbors of i .

Using echo chambers to study multi-polarization



Iteration: 1

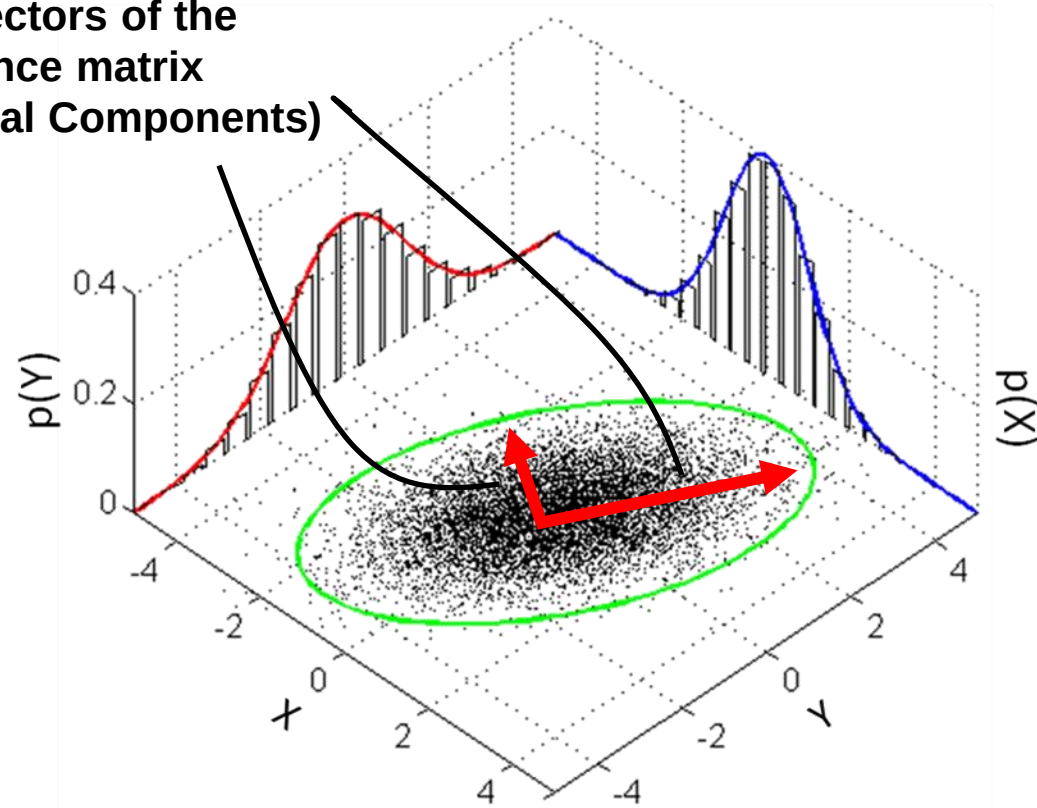
Martin-Gutierrez et al. *Chaos, Solitons & Fractals* **169**, 113244 (2023).

Using echo chambers to study multi-polarization

Eigenvectors of the covariance matrix
(Principal Components)

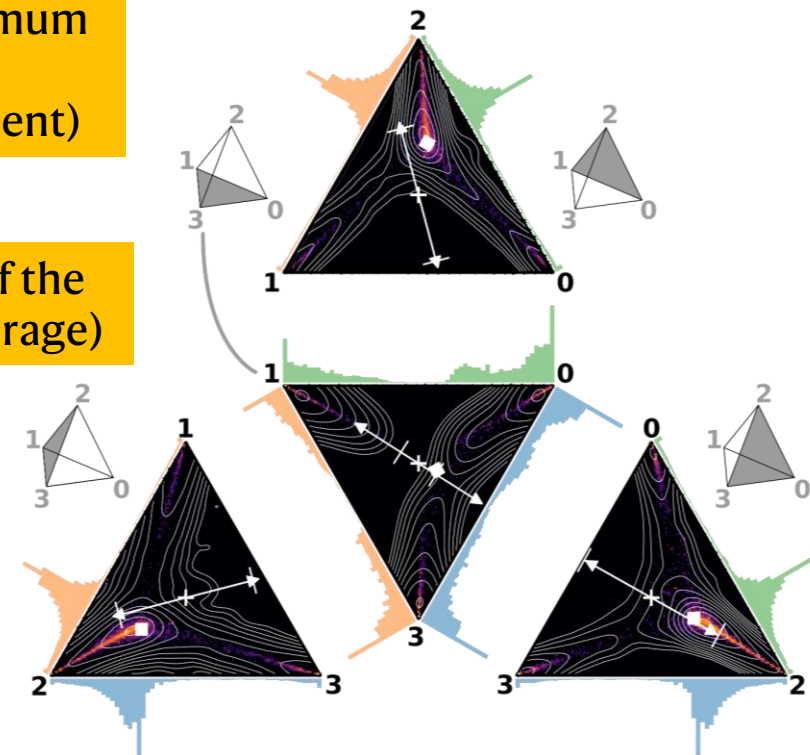
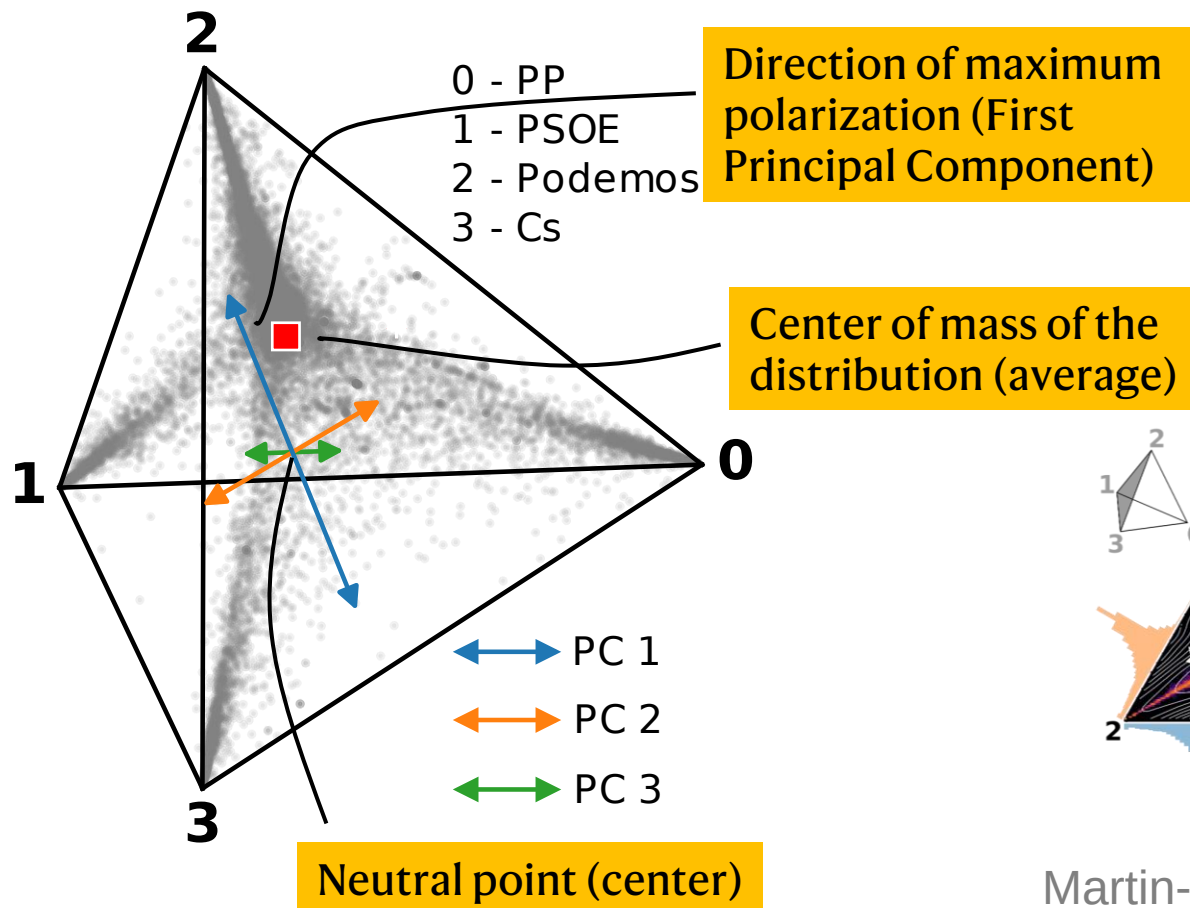
Principal Component Analysis
(PCA)

$$\text{Cov}[\vec{X}, \vec{X}] = \begin{bmatrix} 1 & 3/5 \\ 3/5 & 2 \end{bmatrix}$$



Martin-Gutierrez et al. *Chaos, Solitons & Fractals* **169**, 113244 (2023).

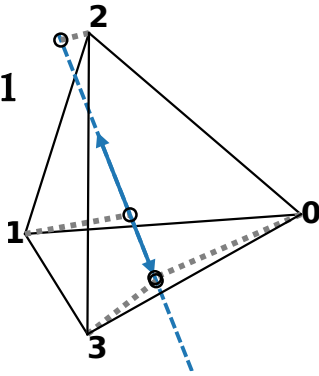
Quadripolar system: Spanish elections 2015



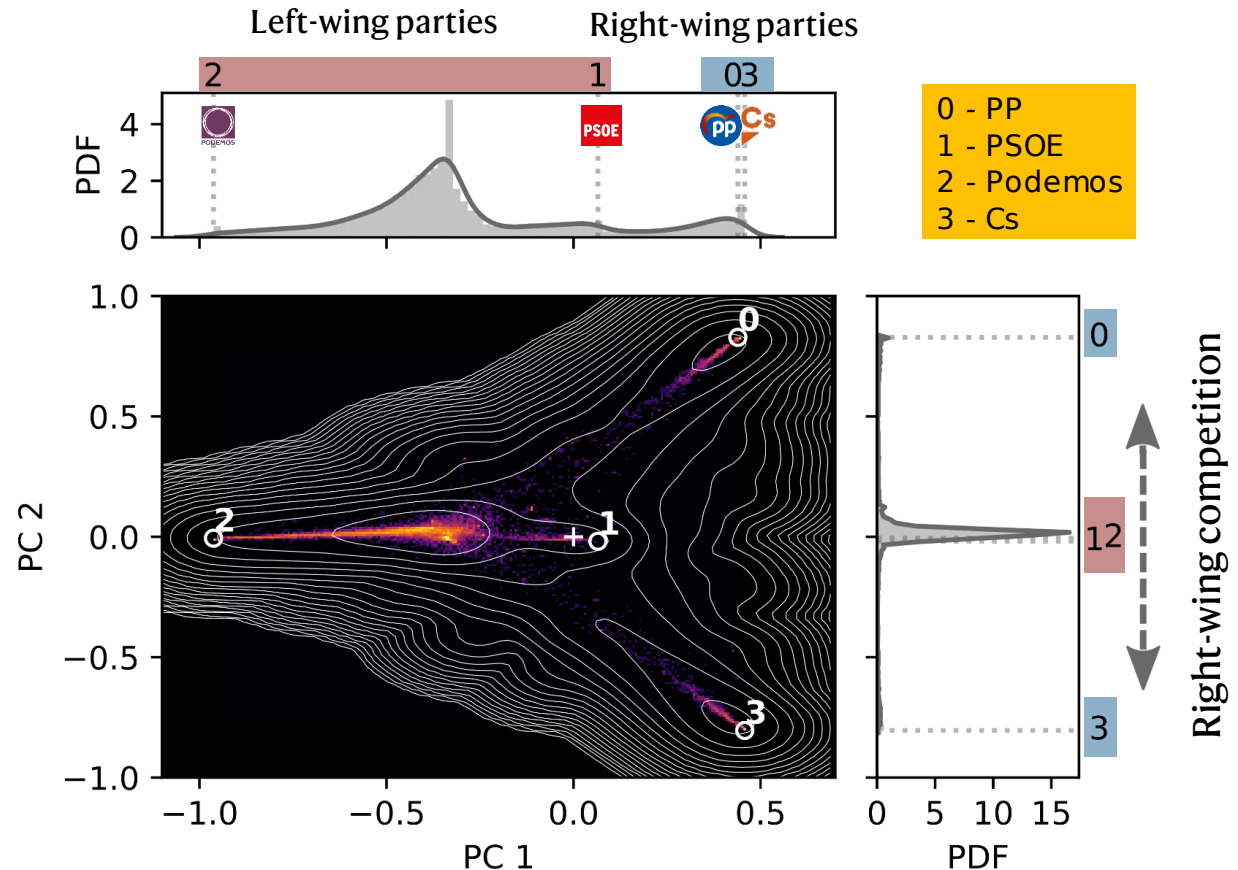
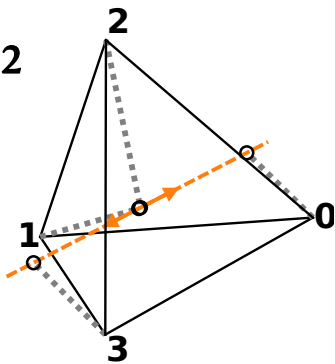
Martin-Gutierrez et al. *Chaos, Solitons & Fractals* **169**, 113244 (2023).

Quadripolar system: Spanish elections 2015

Principal Component 1



Principal Component 2



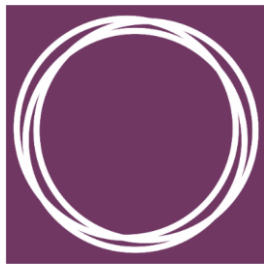
Pentapolar system: Spanish elections of April 2019



0 - Christian conservatism



1 - Social democratic



PODEMOS

2 - Left-wing populism

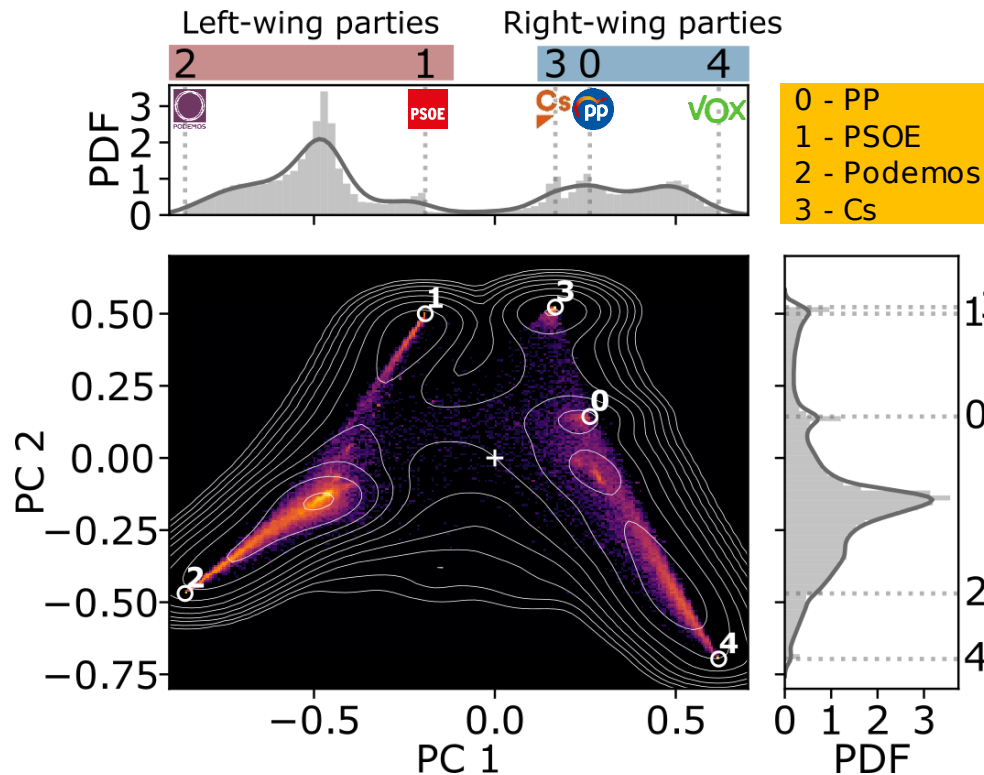


3 - Liberalism



4 - Right-wing populism

Pentapolar system: Spanish elections 2019



CIS Centro de Investigaciones Sociológicas

National survey on parties' ideology

